



Sparse Stochastic Optimal Control under Control-Dependent Diffusion and Time-Varying Dynamics

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Abstract

Sparse stochastic optimal control with control-dependent diffusion arises in robotics, energy systems, and other settings where actuation changes not only the drift but also the noise intensity. The corresponding Hamilton–Jacobi–Bellman (HJB) equations contain nonconvex ℓ_0 penalties, nonsmooth scaled ℓ_1 relaxations, and Hamiltonians that may be discontinuous after minimization over the control. We develop a relaxed viscosity formulation based on the upper and lower semicontinuous Hamiltonian envelopes H^* and H_* . Subsolution tests use H^* , while supersolution tests use H_* ; this asymmetric use of the envelopes gives stable viscosity inequalities when the ℓ_0 switching cost creates discontinuities in the minimized Hamiltonian. Under uniform-in-control local modulus assumptions, and for radial model classes that imply these assumptions, we prove a comparison principle and identify the dynamic-programming value function as the unique viscosity solution in the prescribed growth class. We also characterize the structural regimes in which a scaled ℓ_1 problem exactly recovers the ℓ_0 -optimal policy. Exact recovery is not claimed for general control-dependent diffusion: outside decoupled or diffusion-insensitive regimes, diffusion–Hessian coupling can shift the switching surface, and counterexamples and perturbation bounds describe the resulting policy mismatch. Finally, we analyze monotone semi–Lagrangian approximations under explicit consistency, control-net, and stencil-positivity conditions. Numerical experiments on representative control-dependent-noise benchmarks illustrate the predicted convergence trends, bang–off–bang switching, and sparsity patterns.

Keywords Sparse control · Stochastic optimal control · Control-dependent diffusion · Hamilton–Jacobi–Bellman equations · Viscosity solutions

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1 Introduction

Sparse control promotes intervals of actuator inactivity and is used to reduce wear, energy consumption, and communication load. A standard formulation penalizes the number of active control components through an ℓ_0 running cost, also known as maximum hands-off



control [1]. In deterministic systems, the relationship between ℓ_0 costs and convex ℓ_1 formulations is well understood under structural assumptions such as normality, bang–bang optimality, or suitable threshold conditions [2–4]. These ideas have been applied to scheduling, minimum-attention control, and networked systems [5].

The stochastic case becomes more delicate when diffusion depends on the control. Such dependence appears, for example, in robotic systems with actuation-dependent friction, power systems with control-dependent uncertainty, and flow or turbulence models in which the applied input changes the effective noise level. Two difficulties then arise simultaneously. First, the ℓ_0 term is discontinuous at zero control, and minimization over the control can therefore produce a discontinuous Hamiltonian. Second, when the diffusion matrix depends on the control, the Hessian term in the HJB operator also enters the switching decision. Consequently, minimizers may change discontinuously under perturbations of (t, x) , the test gradient, or the test Hessian. Classical comparison arguments for continuous Hamiltonians do not directly apply.

Most existing sparse-control results either treat deterministic dynamics or assume control-independent diffusion, where the stochastic HJB equation fits more directly into the standard viscosity framework [6, 7]. The present paper studies the finite-dimensional stochastic problem in which sparsity and control-dependent diffusion occur together. The aim is not to assert a general convex-relaxation principle for all control-dependent diffusions, but to separate the HJB-level viscosity theory from the additional structural assumptions required for exact ℓ_0 – ℓ_1 recovery.

The main results are summarized as follows.

R1. Hamiltonian-envelope viscosity formulation. We define the minimized Hamiltonian using the control-dependent drift, diffusion, and sparse running cost, and formulate viscosity inequalities with its semicontinuous envelopes H^* and H_* . The upper envelope H^* is used for subsolutions and the lower envelope H_* for supersolutions. This is the point at which the analysis departs from the classical continuous-Hamiltonian viscosity theory. When the minimized Hamiltonian is continuous, the formulation reduces to the standard viscosity definition.

R2. Comparison and well-posedness. Under uniform-in-control local continuity assumptions on the coefficients and costs, we prove a comparison principle for relaxed viscosity sub- and supersolutions. We also identify radial and separable diffusion classes for which the required modulus assumptions can be verified. Combined with the dynamic programming principle and a Perron argument, this yields uniqueness and identifies the value function with the viscosity solution.

R3. Measurability and admissibility of sparse costs. Because $1_{\{u_i \neq 0\}}$ is discontinuous, we explicitly verify measurability and integrability of the ℓ_0 running cost under progressively measurable controls. Strong well-posedness of the controlled SDE is justified by Lipschitz and growth assumptions in the state variable, uniformly over the compact control set.

R4. Boundary of ℓ_0 – ℓ_1 exact recovery. In decoupled regimes, such as box-constrained control-affine drift with diffusion independent of the switching variable, the scaled ℓ_1 formulation recovers the same threshold policy as the ℓ_0 problem. If diffusion depends on the same control component, however, the Hessian term can shift the switching surface, so exact recovery is no longer automatic. Counterexamples and persistence bounds quantify this boundary.

R5. Monotone approximation and numerical evidence. We prove convergence of monotone semi-Lagrangian schemes under relaxed consistency, control-net refinement, and explicit stencil-positivity conditions. The numerical experiments report representative finite-dimensional benchmarks and illustrate convergence trends, bang-off-bang behavior, and sparsity preservation within the tested parameter ranges.

These results connect viscosity theory for fully nonlinear HJB equations [8–10] with monotone semi-Lagrangian discretisation methods [11, 12]. They also refine deterministic ℓ_0 – ℓ_1 sparse-control equivalence results [2–5] by identifying which mechanisms persist in stochastic HJB problems with control-dependent diffusion and which require additional structural assumptions.

The analysis is restricted to finite-dimensional Itô diffusions. A variant with separate controls in the drift and diffusion, for instance $b(t, x, u)$ and $\sigma(t, x, v)$, can be treated under analogous compactness, measurability, and uniform-modulus assumptions by minimizing the Hamiltonian over the product control set. Infinite-dimensional stochastic evolution equations would require additional semigroup or mild-viscosity techniques and are therefore left as a possible extension rather than treated as a result of the present paper.

2 Problem Formulation

We consider stochastic control systems with nonlinear, time-varying dynamics and *control-dependent* diffusion, governed by the Itô SDE

$$dx(t) = b(t, x(t), u(t)) dt + \sigma(t, x(t), u(t)) dW(t), \quad (1)$$

where $x(t) \in \mathbb{R}^n$ is the state, $u(t) \in U \subset \mathbb{R}^d$ is the control, and $W(t) \in \mathbb{R}^{m_W}$ is an m_W -dimensional standard Brownian motion. The diffusion coefficient is a map

$$\sigma : [0, T] \times \mathbb{R}^n \times U \rightarrow \mathbb{R}^{n \times m_W},$$

and we write

$$a(t, x, u) := \sigma(t, x, u)\sigma(t, x, u)^\top \in \mathbb{S}_+^n$$

for the diffusion covariance matrix. The process is defined on a complete filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \geq 0}, \mathbb{P})$ satisfying the usual conditions. Unlike control-independent formulations, the diffusion term in Eq. 1 changes with the applied control and can therefore affect both the stochastic fluctuations and the switching structure of the optimal policy.

2.1 Control Objective and Admissible Controls

The performance criterion is the sparse stochastic cost functional

$$J(u) = \mathbb{E} \left[\int_0^T \left(f_0(x(t)) + \sum_{i=1}^d f_i(x(t)) 1_{\{u_i(t) \neq 0\}} \right) dt + g(x(T)) \right], \quad (2)$$

where $f_0, f_i, g : \mathbb{R}^n \rightarrow \mathbb{R}_+$ are continuous and nonnegative. The indicator $1_{\{u_i(t) \neq 0\}}$ is the ℓ_0 contribution of the i -th control component and charges a positive cost whenever that component is active.

The admissible control class is

$$\mathcal{U}[0, T] := \left\{ u : [0, T] \times \Omega \rightarrow U \mid \begin{array}{l} u \text{ is progressively measurable with respect to } \{\mathcal{F}_t\}, \\ \mathbb{E} \int_0^T \|u(t)\|^2 dt < \infty \end{array} \right\}.$$

Here $U \subset \mathbb{R}^d$ is compact, convex, and contains the origin. Under the assumptions below, every $u(\cdot) \in \mathcal{U}[0, T]$ generates a unique strong solution of Eq. 1.

Lemma 1 (Well-posedness of the controlled SDE and of the sparse cost) *Assume (A1)–(A4). For every $u \in \mathcal{U}[0, T]$, the processes $b(t, x, u(t))$ and $\sigma(t, x, u(t))$ are progressively measurable in t and satisfy the standard Lipschitz and linear-growth bounds in x , uniformly in the control. Hence Eq. 1 admits a pathwise unique strong solution X^u on $[0, T]$. Moreover, the sparse cost functional (2) is measurable and finite whenever the polynomial moment order required by (A4) is finite along X^u .*

Proof Since u is progressively measurable and b, σ are Borel in (t, u) and continuous in x , the compositions $b(t, x, u(t))$ and $\sigma(t, x, u(t))$ are progressively measurable for each fixed x . Assumptions (A1)–(A2) give the usual state-Lipschitz and linear-growth estimates uniformly over $u \in U$. The standard strong existence and pathwise uniqueness theorem for Itô SDEs therefore applies to Eq. 1. The resulting solution satisfies polynomial moment bounds of the form

$$\mathbb{E} \sup_{0 \leq t \leq T} \|X^u(t)\|^r < \infty$$

for every finite r covered by the growth assumptions.

For each component, u_i is progressively measurable, so $\{(t, \omega) : u_i(t, \omega) \neq 0\}$ is progressively measurable and $1_{\{u_i(t) \neq 0\}}$ is progressively measurable. Since X^u is adapted with continuous paths, $f_i(X^u(t))$ and $g(X^u(T))$ are measurable. Finally, $0 \leq 1_{\{u_i(t) \neq 0\}} \leq 1$, and (A4) bounds the running and terminal costs by a polynomial in $\|X^u\|$. Thus $J(u)$ is well defined. $\square$

2.2 Standing Assumptions and Analytical Challenges

We impose the following structural conditions:

(A1) **Regularity in the state variable and measurability in the control.** There exists $L > 0$ such that, for all $t \in [0, T]$, $x, x' \in \mathbb{R}^n$, and $u \in U$,

$$\|b(t, x, u) - b(t, x', u)\| + \|\sigma(t, x, u) - \sigma(t, x', u)\| \leq L\|x - x'\|.$$

Moreover, b and σ are Borel measurable in (t, u) and continuous in (t, u) uniformly on compact subsets of the state space. The Lipschitz condition is imposed in x , uniformly over u . No Lipschitz regularity in the control variable is needed for strong well-posedness of Eq. 1 under a fixed progressively measurable control. Control-side continuity is used later for Hamiltonian stability, comparison, and control-net approximation.

(A2) **Linear growth.** There exists $C > 0$ such that, for all $(t, x, u) \in [0, T] \times \mathbb{R}^n \times U$,

$$\|b(t, x, u)\| + \|\sigma(t, x, u)\| \leq C(1 + \|x\|).$$

(A3) **Control set.** $U \subset \mathbb{R}^d$ is compact, convex, and contains 0. For the later sparsity-relaxation analysis, define

$$U_i^{\max} := \sup_{u \in U} |u_i| < \infty.$$

(A4) **Cost regularity.** The functions f_0, f_i, g are continuous, nonnegative, and of at most polynomial growth: there exist $q \geq 0$ and $C_f > 0$ such that

$$|f_0(x)| + \sum_{i=1}^d |f_i(x)| + |g(x)| \leq C_f(1 + \|x\|^q), \quad x \in \mathbb{R}^n.$$

Remark 1 (Role of the assumptions) Conditions (A1)–(A2) give strong well-posedness of the controlled SDE for each progressively measurable control, as stated in Lemma 1. The discontinuity of the sparse running cost does not prevent the cost functional from being measurable, because $1_{\{u_i \neq 0\}}$ is measurable for progressively measurable u_i . The discontinuity becomes critical only after pointwise minimization in the HJB equation, where it may make the minimized Hamiltonian discontinuous. This is why the viscosity formulation in Section 3 uses the upper and lower Hamiltonian envelopes H^* and H_* .

When σ depends on the control, the HJB operator remains degenerate elliptic: for fixed u , the map

$$M \mapsto \frac{1}{2} \text{Tr}(\sigma(t, x, u) \sigma(t, x, u)^\top M)$$

is linear and monotone in the semidefinite order, and taking the infimum over $u \in U$ preserves ellipticity. The main issue for comparison is therefore not ellipticity itself, but the possible lack of continuity of the minimized Hamiltonian and the instability of minimizers under perturbations of (t, x, p, M) . Standard measurable-selection arguments enter later when near minimizers are selected in the dynamic-programming argument; they are not needed merely to define the sparse cost in Eq. 2. Structural assumptions for exact ℓ_0 – ℓ_1 recovery are stated separately in Section 4.

Proposition 2 (Separate drift and diffusion controls) Let $U_b \subset \mathbb{R}^{d_b}$ and $U_\sigma \subset \mathbb{R}^{d_\sigma}$ be compact control sets, and consider the controlled diffusion

$$dX_s = b(s, X_s, u_s) ds + \sigma(s, X_s, v_s) dW_s, \quad u_s \in U_b, \quad v_s \in U_\sigma.$$

Assume that b and σ are Borel measurable in their control variables and satisfy, uniformly over $(u, v) \in U_b \times U_\sigma$, the state-Lipschitz and linear-growth bounds

$$\|b(t, x, u) - b(t, x', u)\| + \|\sigma(t, x, v) - \sigma(t, x', v)\| \leq L\|x - x'\|,$$

and

$$\|b(t, x, u)\| + \|\sigma(t, x, v)\| \leq C(1 + \|x\|).$$

Then, for every progressively measurable pair (u, v) , the SDE has a unique strong solution. If the running cost has the sparse form

$$F(x, u, v) = f_0(x) + \sum_{i \in I_b} f_i(x) 1_{\{u_i \neq 0\}} + \sum_{j \in I_\sigma} \tilde{f}_j(x) 1_{\{v_j \neq 0\}},$$

with continuous nonnegative coefficients of polynomial growth, then the corresponding cost functional is measurable and finite under the same moment-growth condition as in Lemma 1. The case in which sparsity is imposed only on the drift control is obtained by taking $I_\sigma = \emptyset$.

The associated pointwise Hamiltonian is

$$H(t, x, p, M) = \inf_{(u, v) \in U_b \times U_\sigma} \left\{ \langle b(t, x, u), p \rangle + \frac{1}{2} \text{Tr}(\sigma(t, x, v) \sigma(t, x, v)^\top M) + F(x, u, v) \right\}.$$

If the structural continuity, growth, and modulus assumptions used in the comparison theorem are imposed uniformly on this product-control Hamiltonian, then the relaxed-viscosity formulation carries over with the semicontinuous envelopes

$$H^*(z) = \limsup_{z' \rightarrow z} H(z'), \quad H_*(z) = \liminf_{z' \rightarrow z} H(z'), \quad z = (t, x, p, M),$$

and the same comparison argument applies.

Proof For fixed progressively measurable controls u and v , the random coefficients

$$(s, \omega, x) \mapsto b(s, x, u_s(\omega)), \quad (s, \omega, x) \mapsto \sigma(s, x, v_s(\omega))$$

are progressively measurable in (s, ω) and satisfy the usual state-Lipschitz and linear-growth bounds with deterministic constants L and C . The standard strong well-posedness theorem for Itô equations with progressively measurable random coefficients therefore gives existence and pathwise uniqueness. The polynomial moment estimate follows from the same growth bound.

The sparse indicators $1_{\{u_i \neq 0\}}$ and $1_{\{v_j \neq 0\}}$ are progressively measurable whenever the corresponding control components are progressively measurable. Combined with the polynomial-growth bound on the coefficients $f_i, \tilde{f}_j$, this proves measurability and finiteness of the cost functional under the same moment estimate.

For the HJB part, replacing U by the compact product set $U_b \times U_\sigma$ changes only the domain of pointwise minimization. For each fixed (u, v) , the second-order term is monotone in M in the semidefinite order; the infimum over the product set preserves degenerate ellipticity. If the comparison hypotheses used for the original Hamiltonian hold uniformly for the product-control Hamiltonian, the relaxed subsolution and supersolution definitions using H^* and H_* are unchanged. The doubling-of-variables comparison argument therefore applies to the product-control formulation under those uniform hypotheses. $\square$

Remark 2 (Scope of the product-control extension) Proposition 2 addresses well-posedness, cost measurability, and the relaxed-HJB formulation for separate drift and diffusion controls. It does not assert that the exact ℓ_0 – ℓ_1 recovery theory automatically transfers to every product-control model. Exact recovery depends on the pointwise structure of the minimized Hamiltonian. If the sparse switching variable enters only through a decoupled linear control channel, the exactness conditions of Section 4 may still apply. If it also enters the diffusion-Hessian term, then diffusion–control coupling can destroy the common bang–off–bang selector, and only the weak-coupling or value-gap statements of Section 4 remain applicable.

3 Generalized HJB Equation

We derive the Hamilton–Jacobi–Bellman (HJB) equation associated with Eqs. 1–2 when the diffusion coefficient depends on the control and the running cost contains a discontinuous ℓ_0 term. The minimized Hamiltonian may fail to be continuous even when the coefficients are continuous before minimization. We therefore use a relaxed viscosity formulation based on the upper and lower semicontinuous envelopes of the minimized Hamiltonian.

3.1 Dynamic Programming and the Relaxed Hamiltonian

Under Assumptions (A1)–(A4), the value function is

$$V(t, x) := \inf_{u \in \mathcal{U}[t, T]} \mathbb{E}^{t, x} \left[\int_t^T \phi(X_s^{t, x; u}, u_s) ds + g(X_T^{t, x; u}) \right], \quad \phi(x, u) := f_0(x) + \sum_{i=1}^d f_i(x) |u_i|_0.$$

By Lemma 1, the controlled SDE and the sparse cost are well defined for every strict progressively measurable control $u \in \mathcal{U}[t, T]$.

Lemma 3 (Dynamic programming principle) *Let (A1)–(A4) hold, and let $U \subset \mathbb{R}^d$ be compact. For $t \in [0, T]$, let $\mathcal{U}[t, T]$ be the class of U -valued progressively measurable controls on $[t, T]$, and let $X^{t, x; u}$ be the unique strong solution of Eq. 1 starting from x at time t . Define*

$$J(t, x; u) := \mathbb{E}^{t, x} \left[\int_t^T \phi(X_s^{t, x; u}, u_s) ds + g(X_T^{t, x; u}) \right], \quad V(t, x) := \inf_{u \in \mathcal{U}[t, T]} J(t, x; u).$$

Then the dynamic-programming identity holds in the standard universally measurable sense: for every bounded stopping time τ taking values in $[t, T]$,

$$V(t, x) = \inf_{u \in \mathcal{U}[t, T]} \mathbb{E}^{t, x} \left[\int_t^\tau \phi(X_s^{t, x; u}, u_s) ds + V(\tau, X_\tau^{t, x; u}) \right]. \quad (3)$$

In particular, Eq. 3 holds for deterministic $\tau = s \in [t, T]$. The value function and the sparse cost are defined on strict progressively measurable controls. The proof uses ε -optimal universally measurable continuation selectors, not the existence of a strict optimal control.

Proof Fix (t, x) and write $X^u := X^{t, x; u}$. By Lemma 1, the payoff $\phi(X_s^u, u_s)$ is progressively measurable and integrable under the stated polynomial-growth condition. The discontinuity of $|u_i|_0$ creates no measurability obstruction at the level of the strict-control cost; its effect appears later through the regularity of the minimized Hamiltonian.

For the inequality “ $\geq$ ”, let $u \in \mathcal{U}[t, T]$ be arbitrary. By the tower property and the strong Markov property of the controlled diffusion,

$$J(t, x; u) = \mathbb{E} \left[\int_t^\tau \phi(X_s^u, u_s) ds + J(\tau, X_\tau^u; u^\tau) \right] \geq \mathbb{E} \left[\int_t^\tau \phi(X_s^u, u_s) ds + V(\tau, X_\tau^u) \right],$$

where u^τ denotes the shifted post- τ control. Taking the infimum over u gives the first inequality.

For the reverse inequality, fix $\varepsilon > 0$. We use the standard measurable-selection form of the dynamic-programming principle for Borel controlled diffusions with compact action space; see the lower-semi-analytic formulation of stochastic dynamic programming and universally measurable ε -optimal selectors in [9, 13, 14]. In this formulation, V may be taken lower semianalytic, and there exists a universally measurable family of continuation controls

$$(s, y) \mapsto v_{s, y}^\varepsilon \in \mathcal{U}[s, T]$$

such that

$$J(s, y; v_{s, y}^\varepsilon) \leq V(s, y) + \varepsilon$$

along the stopped states (τ, X_τ^u) , up to $\mathbb{P}^{t, x; u}$ -null sets. This is the only selection input used here; existence of an exact strict optimal control is not assumed.

Given $u \in \mathcal{U}[t, T]$, define the concatenated control

$$(u \otimes_\tau v^\varepsilon)_r := u_r 1_{\{r < \tau\}} + v_{\tau, X_\tau^u}^\varepsilon(r) 1_{\{r \geq \tau\}}, \quad r \in [t, T].$$

By the standard pasting lemma for progressively measurable controls [13], we have $u \otimes_\tau v^\varepsilon \in \mathcal{U}[t, T]$. Hence

$$J(t, x; u \otimes_\tau v^\varepsilon) \leq \mathbb{E} \left[\int_t^\tau \phi(X_s^u, u_s) ds + V(\tau, X_\tau^u) + \varepsilon \right].$$

Taking the infimum over u and then letting $\varepsilon \downarrow 0$ gives the reverse inequality. $\square$

A formal Itô–DPP calculation leads to the HJB equation

$$-\partial_t V(t, x) + \inf_{u \in U} \left\{ \langle b(t, x, u), \nabla_x V(t, x) \rangle + \frac{1}{2} \text{Tr}(\sigma \sigma^\top(t, x, u) D_x^2 V(t, x)) + \phi(x, u) \right\} = 0, \quad (4)$$

with terminal condition $V(T, x) = g(x)$. Define, for $p \in \mathbb{R}^n$ and $M \in \mathbb{S}^n$,

$$H(t, x, p, M) := \inf_{u \in U} \left\{ \langle b(t, x, u), p \rangle + \frac{1}{2} \text{Tr}(\sigma \sigma^\top(t, x, u) M) + \phi(x, u) \right\}.$$

For fixed u , the map

$$M \mapsto \frac{1}{2} \text{Tr}(\sigma \sigma^\top M)$$

is linear and monotone in the semidefinite order. Therefore the minimized operator is degenerate elliptic. The difficulty is different: because $\phi(x, u)$ contains $|u_i|_0$, the minimization over u can make H discontinuous in (t, x, p, M) .

We use the following semicontinuous envelopes of the minimized Hamiltonian. For $z = (t, x, p, M)$, set

$$H^*(z) := \limsup_{z' \rightarrow z} H(z'), \quad H_*(z) := \liminf_{z' \rightarrow z} H(z').$$

The envelopes do not smooth the control cost and do not by themselves prove comparison. They record the upper and lower limiting Hamiltonians that are stable under perturbations of the test jet. The comparison theorem below still requires a uniform modulus condition, or a structural model class implying it.

Definition 1 (Relaxed viscosity solution) Let V be locally bounded, and define

$$V^*(t, x) := \limsup_{(s, y) \rightarrow (t, x)} V(s, y), \quad V_*(t, x) := \liminf_{(s, y) \rightarrow (t, x)} V(s, y).$$

Then V is a relaxed subsolution of Eq. 4 if, for every (t, x) and every $(q, p, M) \in \mathcal{P}^{2,+} V^*(t, x)$,

$$-q + H^*(t, x, p, M) \leq 0.$$

It is a relaxed supersolution if, for every (t, x) and every $(q, p, M) \in \mathcal{P}^{2,-} V_*(t, x)$,

$$-q + H_*(t, x, p, M) \geq 0.$$

If both conditions hold and $V^*(T, \cdot) \leq g \leq V_*(T, \cdot)$, then V is a relaxed viscosity solution of Eq. 4.

When H is continuous, $H^* = H_* = H$, and Definition 1 reduces to the standard viscosity definition. Thus the present formulation is a conservative extension of the classical one to the discontinuous Hamiltonian generated by ℓ_0 sparsity and control-dependent diffusion.

3.2 Comparison Principle

The comparison argument is stated for the diffusion case without jumps. Its main purpose is to control the Hamiltonian gap in a doubling-of-variables proof despite the discontinuity created by the sparse running cost. The following uniform-in-control modulus condition is the operative assumption used in the proof.

Assumption 4 (Uniform-in-control local modulus continuity) Let $U \subset \mathbb{R}^d$ be compact. For every compact $K \subset [0, T] \times \mathbb{R}^n$ there exist moduli of continuity $\omega_{b,K}$, $\omega_{\sigma,K}$, and $\omega_{\phi,K}$, with $\omega_{*,K}(0) = 0$, such that for all $(t, x), (s, y) \in K$ and all $u \in U$,

$$\|b(t, x, u) - b(s, y, u)\| \leq \omega_{b,K}(|t - s| + \|x - y\|),$$

$$\|\sigma(t, x, u) - \sigma(s, y, u)\|_F \leq \omega_{\sigma,K}(|t - s| + \|x - y\|),$$

and

$$|\phi(x, u) - \phi(y, u)| \leq \omega_{\phi,K}(\|x - y\|).$$

Moreover, b , σ , and ϕ have at most polynomial growth in x , uniformly in t and u , and are Borel measurable in u for each fixed (t, x) . The modulus condition is imposed in (t, x) , uniformly over u ; it does not require continuity of the sparse indicator $|u_i|_0$ as a function of u_i . In the locally Lipschitz case one may take $\omega_{*,K}(r) \leq L_K r$.

When the compact set K is fixed, we write ω_b , ω_σ , and ω_ϕ for moduli dominating the corresponding local moduli on K . Since the proof uses the CIL trace estimate, the diffusion modulus ω_σ is the key quantity.

Equivalently, one may impose the modulus condition directly on $a(t, x, u) = \sigma(t, x, u)\sigma(t, x, u)^\top$: for every compact K there exists a modulus $\Omega_{a,K}$ such that

$$\sup_{u \in U} \|a(t, x, u) - a(s, y, u)\|_F \leq \Omega_{a,K}(|t - s| + \|x - y\|), \quad (t, x), (s, y) \in K.$$

Either the σ -level or the a -level formulation is sufficient for the comparison proof.

Lemma 5 (Uniform-in-control CIL trace bound) Assume (A1)–(A4) and Assumption 4. Let $(t_\varepsilon, s_\varepsilon, x_\varepsilon, y_\varepsilon)$ be the maximizers in the doubling-of-variables construction, and let $(X_\varepsilon, Y_\varepsilon)$ be the Ishii matrices. Set

$$\delta_\varepsilon := |t_\varepsilon - s_\varepsilon| + \|x_\varepsilon - y_\varepsilon\|.$$

Then there exists $C > 0$, independent of ε , such that

$$\sup_{u \in U} |\text{Tr}(a(s_\varepsilon, y_\varepsilon, u)Y_\varepsilon - a(t_\varepsilon, x_\varepsilon, u)X_\varepsilon)| \leq \frac{C}{\varepsilon} \omega_\sigma(\delta_\varepsilon)^2 + o(1). \quad (5)$$

In particular, the right-hand side tends to zero whenever $\omega_\sigma(\delta_\varepsilon)^2/\varepsilon \rightarrow 0$.

Proof Ishii's lemma gives

$$\begin{pmatrix} X_\varepsilon & 0 \\ 0 & -Y_\varepsilon \end{pmatrix} \leq \frac{3}{\varepsilon} \begin{pmatrix} I & -I \\ -I & I \end{pmatrix} + o(1)I.$$

With $S = \sigma(t_\varepsilon, x_\varepsilon, u)$ and $T = \sigma(s_\varepsilon, y_\varepsilon, u)$, the CIL trace estimate gives

$$\text{Tr}(SS^\top X_\varepsilon) - \text{Tr}(TT^\top Y_\varepsilon) \leq \frac{3}{\varepsilon} \|S - T\|_F^2 + o(1).$$

Assumption 4 yields $\|S - T\|_F \leq \omega_\sigma(\delta_\varepsilon)$ uniformly in u . Applying the same estimate with the roles of $(t_\varepsilon, x_\varepsilon, X_\varepsilon)$ and $(s_\varepsilon, y_\varepsilon, Y_\varepsilon)$ interchanged gives the absolute-value form Eq. 5. $\square$

Theorem 6 (Generalized comparison principle) Assume (A1)–(A4) and Assumption 4. Let V_1 be an upper semicontinuous relaxed subsolution and V_2 a lower semicontinuous relaxed supersolution of Eq. 4. Assume that both functions have at most polynomial growth in x , uniformly in t , and that $V_1(T, \cdot) \leq g \leq V_2(T, \cdot)$. Then

$$V_1(t, x) \leq V_2(t, x) \quad \text{for all } (t, x) \in [0, T] \times \mathbb{R}^n.$$

Proof Fix $\eta, \mu > 0$ and $\varepsilon \in (0, 1)$. Let $\langle z \rangle = (1 + \|z\|^2)^{1/2}$, and set

$$\Psi(t, s, x, y) := V_1(t, x) - V_2(s, y) - \frac{\|x - y\|^2}{2\varepsilon} - \frac{(t - s)^2}{2\varepsilon} - \eta(\langle x \rangle^{2q} + \langle y \rangle^{2q}) - \frac{\mu}{T - t} - \frac{\mu}{T - s},$$

where q dominates the polynomial growth of V_1 and V_2 . The maximum of Ψ is attained at $(t_\varepsilon, s_\varepsilon, x_\varepsilon, y_\varepsilon)$ with $t_\varepsilon, s_\varepsilon < T$, and the standard penalization argument gives

$$\frac{\|x_\varepsilon - y_\varepsilon\|^2}{\varepsilon} + \frac{|t_\varepsilon - s_\varepsilon|^2}{\varepsilon} \longrightarrow 0 \quad \text{as } \varepsilon, \eta, \mu \downarrow 0. \quad (6)$$

Thus, with

$$\delta_\varepsilon = |t_\varepsilon - s_\varepsilon| + \|x_\varepsilon - y_\varepsilon\|,$$

we have $\delta_\varepsilon \rightarrow 0$ and $\delta_\varepsilon^2/\varepsilon \rightarrow 0$. In the locally Lipschitz case this implies $\omega_\sigma(\delta_\varepsilon)^2/\varepsilon \rightarrow 0$. For a general modulus, this convergence is included in the comparison modulus condition. This is the only point at which the quantitative modulus of the control-dependent diffusion enters the proof.

By Ishii's lemma,

$$(a_\varepsilon, p_\varepsilon, X_\varepsilon) \in \mathcal{P}^{2,+}V_1^*(t_\varepsilon, x_\varepsilon), \quad (b_\varepsilon, p_\varepsilon, Y_\varepsilon) \in \mathcal{P}^{2,-}V_{2*}(s_\varepsilon, y_\varepsilon),$$

with

$$p_\varepsilon = \frac{x_\varepsilon - y_\varepsilon}{\varepsilon} + o(1),$$

$$a_\varepsilon = \frac{t_\varepsilon - s_\varepsilon}{\varepsilon} + \frac{\mu}{(T - t_\varepsilon)^2} + o(1), \quad b_\varepsilon = \frac{t_\varepsilon - s_\varepsilon}{\varepsilon} - \frac{\mu}{(T - s_\varepsilon)^2} + o(1),$$

and the usual matrix inequality. The relaxed subsolution and supersolution inequalities give

$$-a_\varepsilon + H^*(t_\varepsilon, x_\varepsilon, p_\varepsilon, X_\varepsilon) \leq 0, \quad -b_\varepsilon + H_*(s_\varepsilon, y_\varepsilon, p_\varepsilon, Y_\varepsilon) \geq 0.$$

Therefore

$$0 \leq (a_\varepsilon - b_\varepsilon) + H^*(t_\varepsilon, x_\varepsilon, p_\varepsilon, X_\varepsilon) - H_*(s_\varepsilon, y_\varepsilon, p_\varepsilon, Y_\varepsilon). \quad (7)$$

Let

$$F(t, x, p, M; u) := \langle b(t, x, u), p \rangle + \frac{1}{2} \text{Tr}(a(t, x, u)M) + \phi(x, u).$$

By compactness of U and the definitions of H^* and H_* , after passing to nearby points realizing the corresponding limsup and liminf in the envelope definitions, the Hamiltonian gap in Eq. 7 is bounded by the uniform control gap:

$$\begin{aligned} & H^*(t_\varepsilon, x_\varepsilon, p_\varepsilon, X_\varepsilon) - H_*(s_\varepsilon, y_\varepsilon, p_\varepsilon, Y_\varepsilon) \\ & \leq \sup_{u \in U} \{F(t_\varepsilon, x_\varepsilon, p_\varepsilon, X_\varepsilon; u) - F(s_\varepsilon, y_\varepsilon, p_\varepsilon, Y_\varepsilon; u)\} + o(1). \end{aligned}$$

On a compact set containing the maximizing points, uniformly in $u \in U$,

$$\begin{aligned} & |\langle b(t_\varepsilon, x_\varepsilon, u) - b(s_\varepsilon, y_\varepsilon, u), p_\varepsilon \rangle| \\ & \leq \omega_b(\delta_\varepsilon) \left(\frac{\|x_\varepsilon - y_\varepsilon\|}{\varepsilon} + o(1) \right) = o(1), \end{aligned}$$

where the last step follows from the same modulus condition used in the doubling argument. The diffusion part is controlled by Lemma 5. The running cost satisfies

$$|\phi(x_\varepsilon, u) - \phi(y_\varepsilon, u)| \leq \omega_\phi(\|x_\varepsilon - y_\varepsilon\|) = o(1),$$

uniformly in u . Hence the Hamiltonian gap is $o(1)$. Since

$$a_\varepsilon - b_\varepsilon = \mu((T - t_\varepsilon)^{-2} + (T - s_\varepsilon)^{-2}) + o(1),$$

letting $\mu \downarrow 0$ and then $\varepsilon, \eta \downarrow 0$ in Eq. 7 gives $\sup(V_1 - V_2) \leq 0$. Therefore $V_1 \leq V_2$ on $[0, T] \times \mathbb{R}^n$. $\square$

Proposition 7 (*Half-relaxed limits stability*) Assume (A1)–(A4) and Assumption 4. Upper half-relaxed limits of relaxed subsolutions are relaxed subsolutions, and lower half-relaxed limits of relaxed supersolutions are relaxed supersolutions.

Proof Let $\bar{V} = \limsup_{n \rightarrow \infty}^* V^n$ be the upper half-relaxed limit of relaxed subsolutions V^n . Let $\varphi \in C^{1,2}$ touch $\bar{V}$ from above at (t_0, x_0) . The standard half-relaxed-limits selection argument gives points $(t_n, x_n) \rightarrow (t_0, x_0)$ at which $V^n - \varphi$ has a local maximum and

$$-\partial_t \varphi(t_n, x_n) + H^*(t_n, x_n, \nabla \varphi(t_n, x_n), D^2 \varphi(t_n, x_n)) \leq 0.$$

Since H^* is upper semicontinuous by definition, passing to the limit gives the subsolution inequality for $\bar{V}$. The lower half-relaxed limit of supersolutions is treated similarly using local minima and H_* . The terminal condition and polynomial growth bounds pass by semicontinuity. $\square$

3.3 Verifiable Structural Classes

The comparison proof uses Assumption 4. We now record model classes in which this assumption can be verified directly. Radial monotonicity is used here only as part of such verifiable structures. It is not a stand-alone replacement for Assumption 4 unless accompanied by the stated continuity or separability conditions.

Assumption 8 (Radial monotonicity from the origin) For each $(t, x) \in [0, T] \times \mathbb{R}^n$, each unit vector $\xi \in \mathbb{S}^{d-1}$, and each $M \geq 0$, the map

$$r \mapsto \text{Tr}(\sigma \sigma^\top(t, x, r\xi)M)$$

is nondecreasing on every interval on which $r\xi \in U$.

Proposition 9 (*Separable radial structure implies the comparison modulus*) Suppose (A1)–(A4) hold. Assume that, for every compact $K \subset [0, T] \times \mathbb{R}^n$, there exist moduli $\tilde{\omega}_b$ and $\tilde{\omega}_\sigma$ such that

$$\sup_{u \in U} \|b(t, x, u) - b(s, y, u)\| \leq \tilde{\omega}_b(|t - s| + \|x - y\|),$$

and

$$\sup_{u \in U} \|\sigma(t, x, u) - \sigma(s, y, u)\|_F \leq \tilde{\omega}_\sigma(|t - s| + \|x - y\|)$$

for all $(t, x), (s, y) \in K$. If $f_0, \dots, f_d$ are uniformly continuous on compact subsets of $\mathbb{R}^n$, then Assumption 4 holds.

In particular, Assumption 4 holds for the separable radial class

$$\sigma(t, x, u) = a(\|u\|)S(t, x), \quad (8)$$

where S is continuous on compacts and $a : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is locally bounded. If a is nondecreasing, then Assumption 8 also holds.

Proof The first part follows immediately by taking the moduli in Assumption 4 to be the displayed uniform moduli and

$$\omega_\phi(r) := \omega_{f_0}(r) + \sum_{i=1}^d \omega_{f_i}(r),$$

because $|u_i|_0 \leq 1$. For Eq. 8, compactness of U and local boundedness of a give

$$A_{\max} := \sup_{u \in U} |a(\|u\|)| < \infty,$$

and therefore

$$\sup_{u \in U} \|\sigma(t, x, u) - \sigma(s, y, u)\|_F \leq A_{\max} \|S(t, x) - S(s, y)\|_F.$$

Uniform continuity of S on compact sets supplies the required modulus. If a is nondecreasing, then $r \mapsto a(r)^2$ is nondecreasing, and

$$\text{Tr}(\sigma \sigma^\top M) = a(\|u\|)^2 \text{Tr}(SS^\top M)$$

is nondecreasing along rays $u = r\xi$ for $M \geq 0$. □

Remark 3 Typical examples satisfying Proposition 9 include isotropic scaling models of the form

$$\sigma(t, x, u) = \sigma_0(1 + \alpha\|u\|)S(t, x),$$

whenever S is uniformly continuous on compacts. If $\alpha \geq 0$, these models also satisfy radial monotonicity. In the scalar case $n = m_W = 1$, models such as

$$\sigma(t, x, u) = \sigma_0(1 + \beta|u|)S(t, x), \quad \sigma(t, x, u) = \sigma_0\sqrt{1 + \beta u^2}S(t, x),$$

satisfy the same condition for $\beta \geq 0$.

Finite radial superpositions are also covered. For

$$\sigma(t, x, u) = \sum_{k=1}^K a_k(\|P_k u\|)S_k(t, x),$$

with fixed linear maps P_k , locally bounded functions a_k , and maps S_k uniformly continuous on compacts, compactness of U gives

$$A_{k,\max} := \sup_{u \in U} |a_k(\|P_k u\|)| < \infty.$$

Hence

$$\sup_{u \in U} \|\sigma(t, x, u) - \sigma(s, y, u)\|_F \leq \left(\sum_{k=1}^K A_{k,\max} \right) \max_{1 \leq k \leq K} \|S_k(t, x) - S_k(s, y)\|_F.$$

Thus Assumption 4 holds on compacts. If the functions a_k are additionally nondecreasing in the relevant radial variables, the corresponding radial monotonicity condition also holds.

Corollary 10 (*Comparison under structural classes*) Under (A1)–(A4), if b , σ , and the costs satisfy Assumption 4, or if they belong to one of the structural classes in Proposition 9, then Theorem 6 yields the comparison principle.

3.4 Well-posedness, Verification, and Regularity

We now identify the value function with the unique relaxed viscosity solution and record the verification and regularity statements used later. The arguments are standard in structure, but the Hamiltonian envelopes H^* and H_* must be kept throughout because the minimized Hamiltonian may be discontinuous.

3.4.1 Perron Construction and Identification of the Value Function

Let $\mathcal{S}$ be the family of lower semicontinuous relaxed supersolutions w of Eq. 4 with at most polynomial growth and terminal condition $w(T, \cdot) \geq g$. Let $\mathcal{R}$ be the family of upper semicontinuous relaxed subsolutions v with at most polynomial growth and terminal condition $v(T, \cdot) \leq g$. Define

$$\bar{V}(t, x) := \inf_{w \in \mathcal{S}} w(t, x), \quad \underline{V}(t, x) := \sup_{v \in \mathcal{R}} v(t, x),$$

and denote their lower and upper semicontinuous regularizations by $\bar{V}_*$ and $\underline{V}_*$.

Theorem 11 (*Perron envelopes and identification of the value function*) Assume (A1)–(A4) and the comparison principle of Theorem 6. Then:

- (i) $\underline{V}_*$ is a relaxed subsolution of Eq. 4, and $\bar{V}_*$ is a relaxed supersolution.
- (ii) The two envelopes coincide, $\underline{V}_* = \bar{V}_* =: V$, and V is the unique relaxed viscosity solution of Eq. 4 in the prescribed polynomial growth class.
- (iii) If V^{DPP} denotes the dynamic-programming value function, then $V^{\text{DPP}} = V$.

Proof We first note that $\mathcal{S}$ and $\mathcal{R}$ are nonempty. Since the data have at most polynomial growth, one can choose $r \geq q$ and constants $A, B > 0$ large enough so that

$$w^+(t, x) := A(1 + \|x\|^r) + B(T - t), \quad w^-(t, x) := -A(1 + \|x\|^r) - B(T - t)$$

are respectively a relaxed supersolution and a relaxed subsolution, with $w^+(T, \cdot) \geq g$ and $w^-(T, \cdot) \leq g$. This polynomial barrier avoids imposing boundedness of g , which is not assumed in (A4).

The stability of Perron envelopes follows from the half-relaxed-limits argument. If $\varphi \in C^{1,2}$ touches $\bar{V}_*$ from below at (t_0, x_0) , then, by the definition of the lower regularization and of the infimum over $\mathcal{S}$, there exist $w_n \in \mathcal{S}$ and $(t_n, x_n) \rightarrow (t_0, x_0)$ such that $w_n - \varphi$ has a local minimum at (t_n, x_n) and

$$w_n(t_n, x_n) \rightarrow \bar{V}_*(t_0, x_0).$$

Since each w_n is a relaxed supersolution,

$$-\partial_t \varphi(t_n, x_n) + H_*(t_n, x_n, \nabla \varphi(t_n, x_n), D^2 \varphi(t_n, x_n)) \geq 0.$$

Passing to the limit and using the lower semicontinuity encoded by H_* gives the supersolution inequality for $\bar{V}_*$. The proof that $\underline{V}^*$ is a relaxed subsolution is analogous, using local maxima and H^* .

By construction, $\underline{V}^* \leq \bar{V}_*$. The comparison principle gives the reverse inequality, hence

$\underline{V}^* = \bar{V}_* =: V$. This proves existence and uniqueness in the stated growth class.

It remains to identify this solution with the DPP value function. If $v \in \mathcal{R}$, the subsolution property, localization, and Itô's formula applied to smooth approximations of v imply, for every admissible control u and bounded stopping time τ ,

$$v(t, x) \leq \mathbb{E}^{t,x} \left[\int_t^\tau \phi(X_s^{t,x;u}, u_s) ds + v(\tau, X_\tau^{t,x;u}) \right].$$

With $\tau = T$ and $v(T, \cdot) \leq g$, we obtain $v \leq J(\cdot; u)$ for every u , hence $v \leq V^{\text{DPP}}$. Taking the supremum over $v \in \mathcal{R}$ yields $\underline{V}^* \leq V^{\text{DPP}}$. The symmetric argument for $w \in \mathcal{S}$ gives $V^{\text{DPP}} \leq \bar{V}_*$. The equality of the envelopes completes the proof. $\square$

3.4.2 Verification and Bang-off-bang Structure

For $u \in U$, define the unminimized Hamiltonian

$$\mathcal{H}(t, x, u, p, P) := \langle b(t, x, u), p \rangle + \frac{1}{2} \text{Tr}(\sigma \sigma^\top(t, x, u) P) + \phi(x, u).$$

Lemma 12 (Borel selectors and ties) *Let $U \subset \mathbb{R}^d$ be compact and let*

$$Y = [0, T] \times \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{S}^n.$$

Write $y = (t, x, p, P)$ and $\Phi(y, u) = \mathcal{H}(t, x, u, p, P)$. Assume that Φ is a normal integrand: $y \mapsto \Phi(y, u)$ is Borel for every fixed u , and $u \mapsto \Phi(y, u)$ is lower semicontinuous for every fixed y . Then the correspondence

$$\text{Arg}(y) := \arg \min_{u \in U} \Phi(y, u)$$

is nonempty, compact-valued, has a measurable graph, and admits a Borel selector $u^*(y)$.

Proof The space Y is Polish and U is compact metric. Since $\Phi(y, \cdot)$ is lower semicontinuous, the minimum is attained and $\text{Arg}(y)$ is nonempty and compact. If $\{u_k\}_{k \geq 1}$ is a countable dense subset of U , then

$$m(y) := \inf_{u \in U} \Phi(y, u) = \inf_k \Phi(y, u_k)$$

is Borel. Hence

$$\text{Graph}(\text{Arg}) = \{(y, u) : \Phi(y, u) = m(y)\}$$

is Borel in $Y \times U$. The Kuratowski–Ryll–Nardzewski selection theorem gives a Borel selector. $\square$

Remark 4 (Progressive measurability and the ℓ_0 penalty) If $(X_t, p(t), P(t))$ is progressively measurable, then $u_t^* := u^*(t, X_t, p(t), P(t))$ is progressively measurable and hence admissible. The scalar penalty $|u|_0 = 1_{\{u \neq 0\}}$ is lower semicontinuous: at $u = 0$, $\liminf_{v \rightarrow 0} |v|_0 \geq 0 = |0|_0$, while at $u \neq 0$ the function is locally constant. Thus, when the continuous part of $\mathcal{H}$ is continuous in u , the minimization problem admits measurable selectors even in the presence of the ℓ_0 term. This selector statement is an attainability result for pointwise Hamiltonian minimizers; it does not assert existence of a strict optimal control for every initial condition.

Theorem 13 (Verification) *Let V be the unique relaxed viscosity solution from Theorem 11.*

(i) *If $V \in C^{1,2}$ has polynomial growth and a Borel selector*

$$u^*(t, x) \in \arg \min_{u \in U} \mathcal{H}(t, x, u, \nabla_x V(t, x), D_{xx}^2 V(t, x))$$

exists, then the feedback $u_t^ = u^*(t, X_t^*)$ is optimal and $V(t, x) = J(t, x; u^*)$.*

(ii) *If a strict optimal pair $(\bar{X}, \bar{u})$ exists and V admits the required subjet along $\bar{X}$, then for a.e. $r \in [t, T]$ there is $(q(r), p(r), P(r)) \in \mathcal{P}^{2,-} V(r, \bar{X}_r)$ such that*

$$\bar{u}(r) \in \arg \min_{u \in U} \mathcal{H}(r, \bar{X}_r, u, p(r), P(r)) \quad \text{a.s.}$$

Proof For the smooth case, the HJB equation gives

$$-\partial_t V + \mathcal{H}(t, x, u, \nabla V, D^2 V) \geq 0$$

for every $u \in U$. Applying Itô's formula along any admissible control and using localization yields $V(t, x) \leq J(t, x; u)$. If u^* attains the pointwise minimum, equality holds.

For the nonsmooth statement, the assertion is conditional on the existence of an optimal pair and the standard subjet approximation property. Applying the viscosity supersolution inequality along inf-convolutions of V , then passing to a subjet limit, gives

$$0 \leq -q(r) + \mathcal{H}(r, \bar{X}_r, \bar{u}_r, p(r), P(r)).$$

The subsolution inequality gives

$$-q(r) + \inf_{u \in U} \mathcal{H}(r, \bar{X}_r, u, p(r), P(r)) \leq 0.$$

Combining the two inequalities yields the stated pointwise minimization property. $\square$

We next isolate a decoupled case in which the sparse minimizer has an explicit threshold form. This result is useful for interpreting the numerical policies and for comparing ℓ_0 and scaled ℓ_1 formulations.

Assume, in this subsection only, that

$$U = \prod_{i=1}^d [-U_i, U_i], \quad U_i > 0, \quad \lambda_i(x) := \frac{f_i(x)}{U_i}.$$

Assumption 14 (Decoupling) The drift is control affine,

$$b(t, x, u) = b_0(t, x) + B(t, x)u,$$

the diffusion is independent of the switching control,

$$\sigma(t, x, u) \equiv \sigma(t, x),$$

$$\text{and } U = \prod_{i=1}^d [-U_i, U_i].$$

Theorem 15 (Componentwise bang–off–bang rule) Under Assumption 14, the pointwise minimization of $\mathcal{H}$ separates coordinatewise. For a.e. t , an optimal selector may be chosen as

$$\bar{u}_i(t) = \begin{cases} -U_i, & \Psi_i(t) > \lambda_i(\bar{X}_t), \\ 0, & |\Psi_i(t)| \leq \lambda_i(\bar{X}_t), \\ +U_i, & \Psi_i(t) < -\lambda_i(\bar{X}_t), \end{cases} \quad \Psi_i(t) := \langle B_i(t, \bar{X}_t), p(t) \rangle.$$

Proof Under Assumption 14,

$$\mathcal{H}(t, x, u, p, P) = \langle b_0(t, x), p \rangle + \frac{1}{2} \text{Tr}(\sigma \sigma^\top(t, x)P) + \sum_{i=1}^d (\Psi_i(t, x, p)u_i + f_i(x)|u_i|_0).$$

Thus each coordinate solves

$$\min_{u_i \in [-U_i, U_i]} \Psi_i u_i + f_i(x)|u_i|_0.$$

The value at $u_i = 0$ is 0; the best nonzero value is $-U_i|\Psi_i| + f_i(x)$. Hence zero is optimal when $U_i|\Psi_i| \leq f_i(x)$, and otherwise the minimizer is $-U_i \operatorname{sgn}(\Psi_i)$. This is exactly the stated threshold rule. $\square$

Remark 5 (Scope of the bang–off–bang rule) Theorem 15 is a decoupled structural result, not a general statement for control-dependent diffusion. If the same sparse control component enters $\sigma(t, x, u)$, then the Hessian term $\frac{1}{2} \operatorname{Tr}(\sigma \sigma^\top D^2 V)$ may change the switching surface, and the coordinatewise threshold rule can fail. This distinction is the basis for the exact-recovery and counterexample analysis in Section 4.

3.4.3 Regularity of the Value Function

Even without uniform ellipticity, the DPP and SDE stability estimates yield continuity estimates inherited from the data.

Theorem 16 (*A priori Lipschitz–Hölder regularity*) Assume (A1)–(A4) and suppose that $f_0, \dots, f_d, g$ are globally Lipschitz and have at most linear growth. Then there exist constants $L, C > 0$, depending only on the data and on T , such that

$$|V(t, x) - V(t, y)| \leq L\|x - y\|, \quad |V(t, x) - V(s, x)| \leq C(1 + \|x\|)|t - s|^{1/2}.$$

On compact state sets, the second estimate gives the usual local 1/2–Hölder continuity in time.

Proof For the spatial estimate, couple the solutions $X^{x,u}$ and $X^{y,u}$ by using the same Brownian motion and the same admissible control. Assumption (A1), BDG, and Gronwall’s inequality give

$$\mathbb{E} \sup_{r \in [t, T]} \|X_r^{x,u} - X_r^{y,u}\| \leq C\|x - y\|.$$

Since $|u_i|_0 \leq 1$ and the costs are Lipschitz,

$$|J(t, x; u) - J(t, y; u)| \leq C\|x - y\|.$$

Taking the infimum over u gives the spatial Lipschitz bound.

For the time estimate, assume $t < s$. The DPP gives

$$V(t, x) \leq \mathbb{E} \left[\int_t^s \phi(X_r^{x,u}, u_r) dr + V(s, X_s^{x,u}) \right].$$

The growth of the cost and the moment estimate

$$\mathbb{E} \|X_s^{x,u} - x\| \leq C(1 + \|x\|)|s - t|^{1/2},$$

together with the spatial Lipschitz estimate, yield one side of the bound. The reverse inequality is obtained by the usual comparison of controls over the shorter interval and the same SDE estimate. $\square$

Remark 6 (Regularity scope) Under local moduli in Assumption 4 and locally Lipschitz costs, the estimates above have local versions on compact subsets of $[0, T] \times \mathbb{R}^n$. If the diffusion matrix is uniformly elliptic and the coefficients satisfy the standard boundedness and Hölder assumptions, interior Krylov–Safonov estimates yield additional $C^{\alpha/2, \alpha}$ regularity; see [15]. Uniform ellipticity is not used for comparison in Theorem 6; it only strengthens regularity.

3.5 Convergence of a Monotone semi-Lagrangian Scheme

Let $\Delta t > 0$ and $h > 0$ be the time and space steps. We consider a semi-Lagrangian Bellman operator $\mathcal{J}_{\Delta t, h}$ that is monotone, stable, and relaxed-consistent with the envelope HJB. The CFL restriction below is not intended as the only possible discretization strategy; it is a checkable sufficient condition for the positive-stencil scheme analyzed here.

Assumption 17 (Monotonicity, stability, and relaxed consistency) The operators $\mathcal{J}_{\Delta t, h}$ satisfy the following conditions.

- (i) If $V \leq W$, then $\mathcal{J}_{\Delta t, h} V \leq \mathcal{J}_{\Delta t, h} W$.
- (ii) The backward recursion with terminal data g is locally stable on compact computational domains.
- (iii) For every $\varphi \in C^{1,2}$ and every point (t, x) ,

$$\limsup_{\substack{\Delta t, h \rightarrow 0 \\ (t^n, x_i) \rightarrow (t, x)}} \left[\frac{\varphi(x_i) - (\mathcal{J}_{\Delta t, h} \varphi)(x_i)}{\Delta t} - H^*(t, x, \nabla \varphi(x), D^2 \varphi(x)) \right] \leq 0,$$

and

$$\liminf_{\substack{\Delta t, h \rightarrow 0 \\ (t^n, x_i) \rightarrow (t, x)}} \left[\frac{\varphi(x_i) - (\mathcal{J}_{\Delta t, h} \varphi)(x_i)}{\Delta t} - H_*(t, x, \nabla \varphi(x), D^2 \varphi(x)) \right] \geq 0.$$

Let $\{x_i\}_{i \in \mathbb{Z}^n}$ be a Cartesian grid with mesh h , and let $t^n = n\Delta t$. Let I_h be the multilinear interpolant with nonnegative barycentric weights. Let $U_h \subset U$ be a finite control net with mesh $\varepsilon(h) \downarrow 0$ and with $0 \in U_h$. For a bounded grid function V , define

$$\begin{aligned} (\mathcal{J}_{\Delta t, h} V)(x_i) := \min_{u \in U_h} & \left[\Delta t \left(f_0(x_i) + \sum_{j=1}^d f_j(x_i) |u_j|_0 \right) \right. \\ & \left. + \sum_{\ell=1}^{2m_W} \omega_\ell I_h V \left(x_i + \Delta t b(t^n, x_i, u) + \sqrt{\Delta t} \sigma(t^n, x_i, u) \xi_\ell \right) \right], \end{aligned} \quad (9)$$

where $\omega_\ell = 1/(2m_W)$ and $\xi_{k,\pm} = \pm\sqrt{m_W}e_k$. Thus

$$\sum_\ell \omega_\ell \xi_\ell = 0, \quad \sum_\ell \omega_\ell \xi_\ell \xi_\ell^\top = I_{m_W}.$$

The running cost in Eq. 9 is the original ℓ_0 cost, not a weighted ℓ_1 replacement.

Assumption 18 (Positive-footprint CFL condition) On the computational grid region Ω_h , suppose there is $C_{\text{CFL}} > 0$ such that, for all grid nodes $x_i \in \Omega_h$, controls $u \in U_h$, and times t^n ,

$$\|\Delta t b(t^n, x_i, u)\| + \sqrt{\Delta t} \|\sigma(t^n, x_i, u)\|_F \leq C_{\text{CFL}} h.$$

This condition ensures that all footprints in Eq. 9 remain inside the local interpolation stencil with nonnegative interpolation weights.

Remark 7 (Norm convention in the CFL condition) The CFL condition is written with the Frobenius norm of σ . If the operator norm is used instead, a conservative sufficient condition is

$$\|\Delta t b(t^n, x_i, u)\| + \sqrt{\Delta t} \|\sigma(t^n, x_i, u)\|_{2 \rightarrow 2} \sqrt{m_W} \leq C_{\text{CFL}} h.$$

Under the Frobenius convention, the extra factor $\sqrt{m_W}$ is already absorbed by $\|\sigma\|_F$.

Lemma 19 (Checkable sufficient CFL) Assume that

$$B_* := \sup_{\substack{t \in [0, T], x \in \Omega_h \\ u \in U_h}} \|b(t, x, u)\| < \infty, \quad \Sigma_* := \sup_{\substack{t \in [0, T], x \in \Omega_h \\ u \in U_h}} \|\sigma(t, x, u)\|_F < \infty.$$

Fix $\theta \in (0, 1)$ and set $C_{\text{CFL}} = \theta$. If

$$\Delta t \leq \frac{\theta}{2B_*} h, \quad \Delta t \leq \frac{\theta^2}{4\Sigma_*^2} h^2,$$

then Assumption 18 holds.

Proof The first inequality gives $\Delta t B_* \leq \theta h/2$, and the second gives $\sqrt{\Delta t} \Sigma_* \leq \theta h/2$. Adding the two estimates gives the desired footprint bound. $\square$

Remark 8 (Practical verification of the footprint condition) In the numerical experiments, B_* and Σ_* are estimated on the computational domain Ω_h and the discrete control net U_h . The reported mesh pair $(h, \Delta t)$ is then checked against

$$\Delta t \leq \frac{\theta}{2B_*} h, \quad \Delta t \leq \frac{\theta^2}{4\Sigma_*^2} h^2.$$

This records the monotonicity check separately from empirical grid tuning.

Lemma 20 (*Verification of the semi-Lagrangian scheme*) Assume (A1)–(A4), Assumption 4, and the CFL condition in Assumption 18. Then Eq. 9 is monotone and relaxed-consistent. On bounded computational domains with bounded terminal data and bounded running costs, the backward recursion is stable.

Proof Monotonicity follows because I_h is a positive interpolation operator and the cubature weights are nonnegative. Taking the minimum over $u \in U_h$ preserves order.

For consistency, write the continuous part of the unminimized Hamiltonian as

$$\tilde{F}(t, x, p, M; u) = \langle b(t, x, u), p \rangle + \frac{1}{2} \text{Tr}(\sigma \sigma^\top(t, x, u) M).$$

Taylor expansion and the cubature moment identities give, uniformly on compacts,

$$\sum_{\ell} \omega_{\ell} \varphi(x_i + \Delta t b + \sqrt{\Delta t} \sigma \xi_{\ell}) = \varphi(x_i) + \Delta t \tilde{F}(t^n, x_i, \nabla \varphi, D^2 \varphi; u) + O(\Delta t^{3/2}).$$

The interpolation error contributes $O(h^{\alpha})$ before division by Δt . Hence, after division, the local truncation error contains

$$\Delta t^{1/2} + \frac{h^{\alpha}}{\Delta t}.$$

The replacement of U by U_h contributes $\omega_{\text{ctrl}}(\varepsilon(h))$ for the continuous part of the Hamiltonian. The discontinuous ℓ_0 part is retained exactly on U_h and is passed to the limit through H^* and H_* . This gives the relaxed consistency inequalities in Assumption 17 whenever

$$\Delta t^{1/2} + \frac{h^{\alpha}}{\Delta t} + \omega_{\text{ctrl}}(\varepsilon(h)) \rightarrow 0.$$

Stability on bounded computational domains follows by induction from positivity of the interpolation weights and boundedness of the terminal and running costs. $\square$

Remark 9 (Practical role of the CFL restriction) The condition $\Delta t = O(h^2)$ appears because the local positive-footprint stencil must accommodate Brownian displacements of size $\sqrt{\Delta t} \|\sigma\|$. It is a sufficient condition for this particular positive-stencil semi-Lagrangian scheme, not a claim that all monotone approximations of controlled diffusion HJB equations require the same scaling. Wide-stencil, implicit monotone, or probabilistic interpolation schemes may use different consistency–monotonicity tradeoffs. In the numerical section, the reported mesh pairs are checked against the computable bounds in Lemma 19.

Theorem 21 (Convergence) *Assume Assumption 17 and suppose the limiting HJB satisfies the comparison principle in Theorem 6. Then the time-interpolated solutions generated by the backward recursion*

$$V^n = \mathcal{T}_{\Delta t, h} V^{n+1}, \quad V^N = g,$$

converge locally uniformly, along admissible mesh sequences, to the unique relaxed viscosity solution V of Eq. 4.

Proof Define the half-relaxed limits

$$\bar{V} = \limsup^* V^{\Delta t, h}, \quad \underline{V} = \liminf_* V^{\Delta t, h}.$$

Monotonicity, stability, and relaxed consistency imply that $\bar{V}$ is a relaxed subsolution and $\underline{V}$ is a relaxed supersolution of Eq. 4. The terminal condition is inherited from the backward recursion. By comparison, $\bar{V} \leq \underline{V}$, while the definition of half-relaxed limits gives $\bar{V} \geq \underline{V}$. Thus the two limits coincide with the unique solution V , which yields local uniform convergence. $\square$

Corollary 22 (Applicability to the scheme) *Under (A1)–(A4), Assumption 4, and the mesh conditions in Lemma 20, the scheme (9) converges locally uniformly to the unique relaxed viscosity solution of Eq. 4.*

Remark 10 (On ℓ_0 and weighted ℓ_1 discretisations) The operator (9) discretizes the original ℓ_0 running cost. Replacing $|u_j|_0$ by a weighted ℓ_1 term $\lambda_j(x)|u_j|$ generally changes the Hamiltonian and can change the switching surface. An exact identification is available only in special decoupled settings, for example when each coordinate takes values in $\{0, \pm U_j\}$ and $\lambda_j = f_j/U_j$, so that $|u_j| = U_j|u_j|_0$ on the discrete set. Thus the numerical scheme used for the HJB convergence result is an ℓ_0 scheme; scaled ℓ_1 computations are treated separately as relaxation experiments and are not used to prove consistency with the ℓ_0 HJB.

4 ℓ_0 – ℓ_1 Exactness and Its Failure Under Control-Dependent Diffusion

This section identifies the structural boundary between exact scaled ℓ_1 recovery and failure of ℓ_0 – ℓ_1 equivalence. Under a decoupled Hamiltonian structure, a calibrated ℓ_1 penalty has the same minimized Hamiltonian value as the ℓ_0 penalty and admits a common saturated bang–off–bang selector. This is not a general equivalence theorem for arbitrary control-dependent diffusion. When the sparse control also enters the diffusion–Hessian term, the switching surface may change, and exact recovery can fail.

Let

$$U = \prod_{i=1}^d [-U_i, U_i], \quad U_i > 0, \quad \lambda_i(x) := \frac{f_i(x)}{U_i}.$$

The scaled ℓ_1 cost is

$$J_{1,w}(t, x; u) := \mathbb{E}^{t,x} \left[\int_t^T \left(f_0(X_s) + \sum_{i=1}^d \lambda_i(X_s) |u_i(s)| \right) ds + g(X_T) \right]. \quad (10)$$

For every admissible control,

$$\lambda_i(x) |u_i| \leq \lambda_i(x) U_i 1_{\{u_i \neq 0\}} = f_i(x) 1_{\{u_i \neq 0\}},$$

and therefore

$$V_{1,w}(t, x) \leq V_{\ell_0}(t, x). \quad (11)$$

The scaling by U_i is essential. Without it, the comparison may fail unless the control amplitudes are normalized by $|u_i| \leq 1$.

Remark 11 (Attainment and selectors) The equality results below are stated at the Hamiltonian and value-function level. Pointwise minimizers exist because U is compact and the unminimized Hamiltonian is lower semicontinuous in u under the hypotheses used below. Measurable selectors are obtained by Lemma 12. No general claim is made that every relaxed minimizer purifies to a strict optimizer under arbitrary control-dependent diffusion; strict optimal selectors are used only when the stated Hamiltonian minimizers are attainable and measurable.

4.1 Exactness in the Diffusion-insensitive Switching Regime

Theorem 23 (Exactness for diffusion-insensitive switching Hamiltonians) Assume (A1)–(A4), the comparison framework of Section 3.2, and the decoupling condition of Assumption 14: $b(t, x, u) = b_0(t, x) + B(t, x)u$, $\sigma(t, x, u) = \sigma(t, x)$, and $U = \prod_{i=1}^d [-U_i, U_i]$. Let $\lambda_i(x) = f_i(x)/U_i$. Then:

- (a) the minimized ℓ_0 Hamiltonian and the minimized scaled ℓ_1 Hamiltonian have the same value at every (t, x, p, P) ;
- (b) the value functions coincide, $V_{1,w} = V_{\ell_0}$, in the comparison class;
- (c) there exists a common measurable bang–off–bang selector with components

$$u_i^*(t, x, p) = \begin{cases} -U_i, & \Psi_i(t, x, p) > \lambda_i(x), \\ 0, & |\Psi_i(t, x, p)| \leq \lambda_i(x), \\ +U_i, & \Psi_i(t, x, p) < -\lambda_i(x), \end{cases} \quad \Psi_i(t, x, p) := \langle B_i(t, x), p \rangle.$$

If the tie set $\{|\Psi_i| = \lambda_i\}$ has zero time measure along an optimal trajectory, or if a common tie-breaking rule is imposed, the two formulations select the same control a.e. along that trajectory. At threshold points, however, the scaled ℓ_1 stage problem may have additional nonsaturated minimizers; hence the theorem asserts the existence of a common saturated selector, not that every scaled ℓ_1 minimizer is bang–off–bang.

Proof Under the stated decoupling, the u -dependent part of the Hamiltonian separates by coordinates. For the ℓ_0 problem, the i -th stage objective is

$$\Psi_i u_i + f_i 1_{\{u_i \neq 0\}}, \quad u_i \in [-U_i, U_i],$$

whose minimum is

$$\min\{0, f_i - U_i |\Psi_i|\}.$$

A saturated minimizer is 0 when $U_i |\Psi_i| \leq f_i$ and $-U_i \operatorname{sgn}(\Psi_i)$ when $U_i |\Psi_i| > f_i$.

For the scaled ℓ_1 problem, the i -th stage objective is

$$\Psi_i u_i + \lambda_i |u_i|.$$

Its minimum over $[-U_i, U_i]$ is

$$\min\{0, U_i (\lambda_i - |\Psi_i|)\}.$$

Since $\lambda_i = f_i/U_i$, the two minimum values are identical, and the saturated selector above is a minimizer for both problems. Summing over i gives equality of the minimized Hamiltonians.

The comparison principle gives uniqueness for the corresponding relaxed viscosity solutions. Since the terminal condition is the same and the minimized Hamiltonians coincide, $V_{1,w} = V_{\ell_0}$. The measurable saturated selector follows from Lemma 12; the tie statement follows from the preceding one-dimensional minimization. $\square$

Corollary 24 (*Threshold feedback in the exact regime*) Under Theorem 23, any smooth value function or admissible jet selection in the sense of Theorem 13 yields a common threshold feedback for the ℓ_0 and scaled ℓ_1 Hamiltonians. Under stochastic normality, the selector is a.e. unique after the fixed tie-breaking convention.

Remark 12 (Boundary of the exactness statement) Theorem 23 uses three structural ingredients: a box control set, control-affine drift, and diffusion independent of the sparse switching variable. If the diffusion trace depends on the same control component, if the control set is not coordinatewise separable, or if the drift is not affine in the switching variable, the Hamiltonian minimizers may differ. The following results quantify this boundary rather than extend exactness unconditionally.

4.2 Weak Coupling, Margins, and Value-gap Bounds

Assumption 25 (Bounded curvature along relevant jets) Along the trajectories and jets considered in the comparison below, there exists $P_{\max} < \infty$ such that one can choose $(q(t), p(t), P(t))$ from the appropriate viscosity jet with $\|P(t)\| \leq P_{\max}$ for a.e. t .

Proposition 26 (*A sufficient setting for bounded curvature*) Assume (A1)–(A4). Suppose that the diffusion is uniformly elliptic on compact state sets, the coefficients are locally Lipschitz

in (t, x) uniformly in u , and the terminal and running costs are locally semiconcave in x . Then, for every $\tau > 0$ and compact state set $K \subset \mathbb{R}^n$, the value function is locally semiconcave in x on $[0, T - \tau] \times K$. Consequently, the spatial second-order superjets are locally bounded there, and Assumption 25 holds along optimal trajectories that remain in K on $[0, T - \tau]$.

Proof Under the stated parabolic regularity and semiconcavity hypotheses, standard semiconcavity propagation for Bellman equations applies on compact subsets away from the terminal time; see, for example, [10, 16]. Local semiconcavity in x gives an upper quadratic support with a compact-dependent constant. Hence every spatial second-order superjet satisfies $P \leq C_{\tau, K} I$, which yields the required local bound. $\square$

Example 1 (Linear–quadratic data) If

$$b(t, x, u) = A(t)x + B(t)u + a(t), \quad \sigma(t, x, u) = \Sigma(t), \quad \Sigma(t)\Sigma(t)^\top \geq \lambda I_n,$$

and f_0, g are quadratic in x with bounded coefficients on compact time intervals, then the hypotheses of Proposition 26 hold locally. Therefore Assumption 25 is valid along any optimal trajectory that remains in a compact state set before the terminal time.

Proposition 27 (Margin preservation under weak diffusion coupling) Retain (A1)–(A4) and the comparison framework of Section 3.2. Consider a reference ℓ_0 -optimal trajectory $(\bar{X}, \bar{u})$ and choose a jet $(q(t), p(t), P(t))$ along it with $\|P(t)\| \leq P_{\max}$. Let

$$\Psi_i(t) := \langle B_i(t, \bar{X}_t), p(t) \rangle, \quad m_i(t) := |\Psi_i(t) - \lambda_i(\bar{X}_t)|.$$

Let $a = \sigma\sigma^\top$, and define the diffusion–Hessian perturbation caused by changing the i -th channel from 0 to u_i by

$$\Delta_i(u_i; t) := \frac{1}{2} \text{Tr} (a(t, \bar{X}_t, u_i, \bar{u}_{-i}(t)) - a(t, \bar{X}_t, 0, \bar{u}_{-i}(t))) P(t).$$

Assume that, for a.e. t ,

$$m_i(t) \geq \delta_i(t) > 0,$$

and

$$|\Delta_i(u_i; t)| \leq \Gamma_i(t)|u_i|, \quad \Gamma_i(t)U_i < \delta_i(t), \quad u_i \in [-U_i, U_i].$$

Then the channelwise switching decision of the ℓ_0 Hamiltonian is unchanged by the diffusion perturbation along $(\bar{X}, \bar{u})$. If the same margin condition holds uniformly for all states and jets in the comparison domain, then the minimized Hamiltonians coincide in that domain and the equality $V_{1,w} = V_{\ell_0}$ follows by comparison. Along a single trajectory, the conclusion is policy preservation for that trajectory, not a global value-function identity by itself.

Proof For fixed t and i , the unperturbed threshold comparison has slack $m_i(t)$. The diffusion–Hessian perturbation changes the comparison between the inactive value and the saturated active value by at most $\Gamma_i(t)U_i$. If

$$\Gamma_i(t)U_i < \delta_i(t) \leq m_i(t),$$

the sign of the threshold comparison cannot change. Thus the inactive/active decision and the saturated endpoint selected by the Hamiltonian are preserved. The final statement follows only under the stated uniform version, because comparison of value functions requires Hamiltonian agreement on the full domain relevant to the PDE. $\square$

Proposition 28 (Conditional value-gap bound under bounded diffusion coupling) *Assume Assumption 25. Suppose that, on the relevant state–jet domain,*

$$\left| \frac{1}{2} \text{Tr} (a(t, x, u_i, u_{-i}) - a(t, x, 0, u_{-i})) P \right| \leq \kappa |u_i|$$

for every , every $u_i \in [-U_i, U_i]$, and every $\|P\| \leq P_{\max}$. Then the Hamiltonian discrepancy between the ℓ_0 and scaled ℓ_1 switching problems is bounded by

$$\kappa \sum_{i=1}^d U_i.$$

Consequently, for solutions compared on this common domain,

$$0 \leq V_{\ell_0}(t, x) - V_{1,w}(t, x) \leq \kappa \left(\sum_{i=1}^d U_i \right) (T - t),$$

provided the comparison principle applies to both equations and the coupling bound is valid along the relevant viscosity jets.

Proof The pointwise Hamiltonian discrepancy induced by diffusion–control coupling is bounded channelwise by $\kappa|u_i|$ and hence by κU_i on the box. Summing over gives the displayed Hamiltonian bound. The value estimate follows from the standard comparison argument for HJB equations with Hamiltonians differing by a bounded additive term over the time interval $[t, T]$. $\square$

Remark 13 (Diffusion-aware reweighting) If σ depends on the sparse control, one may use a conservative weight

$$\lambda_i^{\text{corr}}(t, x) = \frac{f_i(x)}{U_i} + \Lambda_i(t, x),$$

where Λ_i bounds the channelwise diffusion–Hessian perturbation on the relevant jet set. Such reweighting can preserve a particular switching decision under a margin condition, but it does not restore exact ℓ_0 – ℓ_1 value equality in general.

Proposition 29 (Policy preservation under conservative reweighting) *Let*

$$\lambda_i^{\text{corr}}(t, x) = \lambda_i(x) + \Lambda_i(t, x), \quad \Lambda_i \geq 0.$$

Consider a reference threshold rule with switching function $\Psi_i(t)$ and margin

$$m_i(t) = |\Psi_i(t)| - \lambda_i(\bar{X}_t)$$

along a reference ℓ_0 -optimal trajectory $(\bar{X}, \bar{u})$. If

$$A_i(t, \bar{X}_t) < m_i(t)$$

for a.e. t and all i , then the reweighted scaled ℓ_1 Hamiltonian selects the same saturated control as the reference ℓ_0 Hamiltonian along $(\bar{X}, \bar{u})$. Moreover,

$$0 \leq J_{1,w^{\text{corr}}}(t, x; \bar{u}) - J_{\ell_0}(t, x; \bar{u}) \leq \mathbb{E}^{t,x} \int_t^T \sum_{i=1}^d A_i(s, \bar{X}_s) U_i ds.$$

This is a trajectory-level policy-preservation statement. It becomes a value-function statement only if the preserved selector is also globally optimal for the reweighted problem.

Proof Replacing λ_i by $\lambda_i + A_i$ shifts the threshold by A_i . If this shift is smaller than the margin $m_i(t)$, the ordering of the inactive and saturated active candidates is unchanged. Hence the selected saturated control agrees with the reference ℓ_0 selector along the trajectory. At each active channel, the extra running cost is at most $A_i U_i$; inactive channels contribute zero. Integrating and taking expectation gives the displayed bound. $\square$

4.3 Discrete Operator Bounds and Exact Agreement Under Margins

Let $T_{\ell_0,h}$ and $T_{1,w,h}$ be monotone semi-Lagrangian one-step operators with the same positive interpolation weights and the same control grid. On a bounded computational domain Ω_h , define

$$C_{\Omega_h} := \sup_{x \in \Omega_h} \sum_{i=1}^d (f_i(x) + \lambda_i(x) U_i).$$

Proposition 30 (Operator-level bounds and exact agreement under margins) Assume (A1)–(A4) and Assumption 14. Then:

(a) for any bounded grid function V ,

$$\|T_{\ell_0,h} V - T_{1,w,h} V\|_{L^\infty(\Omega_h)} \leq C_{\Omega_h} \Delta t;$$

(b) if the uniform margin condition in Proposition 27 holds at all grid nodes, controls, and interpolation evaluation points, then the two one-step operators have the same saturated minimizer and the same minimized value at those nodes;

(c) if all interpolation footpoints associated with a node lie in a single cell on which the interpolant is affine in the footpoint, then the one-step switching objective is affine in each control component plus either the ℓ_0 or scaled ℓ_1 penalty. In that case, the threshold comparison reduces to the stagewise analysis in Theorem 23.

Proof For fixed (x, u) , the only direct difference between the two one-step objectives is the penalty term. Since $|u_i| \leq U_i$,

$$|\phi_{e_0}(x, u) - \phi_{1,w}(x, u)| \leq \sum_{i=1}^d (f_i(x) + \lambda_i(x)U_i).$$

Multiplication by Δt and taking minima gives (a). Parts (b) and (c) follow from the same channelwise threshold comparison used in Theorem 23. $\square$

Corollary 31 (*Componentwise search reduction*) *In the decoupled box regime, the saturated threshold rule reduces the inner minimization from enumeration over all ternary active/inactive patterns, of size $O(3^d)$, to d independent scalar comparisons, of size $O(d)$. This computational reduction is exact only under the stated decoupling and margin conditions; otherwise it should be treated as a heuristic or approximate policy-search rule.*

4.4 Empirical Assessment of Near-threshold Events

The stochastic normality condition used to obtain an a.e. unique bang–off–bang selector is a structural assumption, not something that can be certified by finitely many simulations. The numerical role of this subsection is therefore narrower: we measure how often simulated switching scores fall near their activation thresholds.

For $i = 1, \dots, d$, set

$$\Psi_i(t) := \langle B_i(t, \bar{X}(t)), p(t) \rangle, \quad \lambda_i(\bar{X}(t)) := \frac{f_i(\bar{X}(t))}{U_i},$$

where $(\bar{X}, \bar{u})$ is the reference trajectory and $p(t)$ is either an adjoint variable or a viscosity-jet momentum as in Theorem 13. On a grid $\{t_k\}_{k=0}^N$, define

$$\mathcal{T}_i^{(m)} := \left\{ 0 \leq k \leq N : \left| \Psi_i^{(m)}(t_k) - \lambda_i(\bar{X}^{(m)}(t_k)) \right| \leq \varepsilon_{\text{tol}} \right\}.$$

We report the empirical density

$$\hat{\rho}_i := \frac{1}{M(N+1)} \sum_{m=1}^M |\mathcal{T}_i^{(m)}|.$$

Small values of $\hat{\rho}_i$ support the numerical plausibility of a nondegenerate switching surface for the tested model and mesh, but they do not prove that the continuum tie set has Lebesgue measure zero.

Lemma 32 (*Discrete density and time proportion*) *Fix a path for which*

$$Y_i(t) := |\Psi_i(t) - \lambda_i(\bar{X}(t))|$$

is continuous. If the boundary of $\{s : |Y_i(s)| \leq \varepsilon\}$ has Lebesgue measure zero, then

$$\lim_{N \rightarrow \infty} \frac{1}{N+1} \#\{0 \leq k \leq N : |Y_i(t_k)| \leq \varepsilon\} = \frac{1}{T-t} \text{Leb}\{s \in [t, T] : |Y_i(s)| \leq \varepsilon\}.$$

Proof Under the stated boundary condition, the indicator of the set $\{|Y_i| \leq \varepsilon\}$ is Riemann integrable. The assertion is therefore the convergence of uniform Riemann sums. $\square$

Remark 14 (Tolerance and computation of switching scores) For a semi-Lagrangian approximation with spatial order α and time step Δt , a mesh-dependent tolerance may be chosen as

$$\varepsilon_{\text{tol}} = C(h^\alpha + \Delta t^{1/2}).$$

In computations, $\Psi_i(t_k)$ is approximated by

$$\langle B_i(t_k, \bar{X}(t_k)), \nabla_x V_{\Delta t, h}(t_k, \bar{X}(t_k)) \rangle,$$

with the gradient recovered from the local interpolation cell.

Proposition 33 (Monte Carlo concentration for near-threshold density) Fix $\varepsilon > 0$. Let

$$\rho_{i,N}^{(m)} := \frac{1}{N+1} \sum_{k=0}^N 1_{\{|Y_i^{(m)}(t_k)| \leq \varepsilon\}}, \quad \hat{\rho}_{i,N} := \frac{1}{M} \sum_{m=1}^M \rho_{i,N}^{(m)}.$$

Then, for any $\eta > 0$,

$$\mathbb{P}\left(|\hat{\rho}_{i,N} - \mathbb{E}\rho_{i,N}^{(1)}| > \eta\right) \leq 2 \exp(-2M\eta^2).$$

If Y_i is a.s. continuous and the boundary condition in Lemma 32 holds a.s., then $\mathbb{E}\rho_{i,N}^{(1)}$ converges to the expected time proportion of the ε -near-threshold set.

Proof The variables $\rho_{i,N}^{(m)}$ are i.i.d. and take values in $[0, 1]$, so Hoeffding's inequality gives the concentration bound. The limiting statement follows from Lemma 32 and dominated convergence. Figure 1 illustrates a representative switching-score threshold crossing, while Table 1 summarizes the corresponding Monte Carlonear-threshold statistics. $\square$

4.5 Necessity and Sharpness of the Equivalence Assumptions

The hypotheses of Theorem 23 are sharp at the stagewise Hamiltonian level. If the scaling, coordinatewise box structure, or diffusion-insensitive switching condition is removed, one can construct data for which the ℓ_0 and scaled ℓ_1 minimizers differ.

Fix one coordinate and write

$$L_0(u) = \Psi u + f 1_{\{u \neq 0\}}, \quad L_1(u) = \Psi u + \lambda |u|, \quad u \in [-U, U].$$

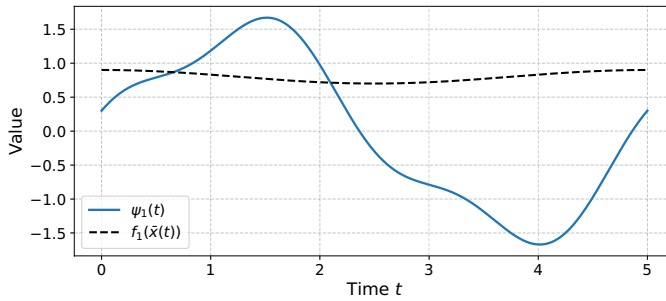


Fig. 1 Switching score $\Psi_1(t)$ and scaled threshold $\lambda_1(\bar{X}(t))$ along a representative trajectory. Switches occur when $|\Psi_1(t)|$ crosses the threshold

Table 1 Near-threshold statistics for $\Psi_i(t)$ over $M = 500$ runs. Density is $\hat{\rho}_i = \frac{1}{M(N+1)} \sum_{m=1}^M |\sigma_i^{(m)}|$

Control dim.	Avg. # pts	Density $\hat{\rho}_i$	Std. dev.
1	3.2	0.0064	1.1
2	4.7	0.0094	1.3
3	0.5	0.0010	0.8

Proposition 34 (Scaled weights are necessary) *In the box, control-affine, diffusion-insensitive setting, if the saturated ℓ_0 and scaled ℓ_1 threshold selectors agree for all slopes Ψ , then $\lambda = f/U$.*

Proof The ℓ_0 objective activates exactly when $|\Psi| > f/U$, while the scaled ℓ_1 objective activates exactly when $|\Psi| > \lambda$. Agreement for all Ψ forces the thresholds to be identical, hence $\lambda = f/U$. $\square$

Proposition 35 (Coordinatewise box separability is necessary) *Let $U = \{u \in \mathbb{R}^d : \|u\|_2 \leq U\}$ with $d \geq 2$. Suppose $f_i = f > 0$ and use $\lambda_i = f/U$. Then there are slopes Ψ for which the stagewise minimizers of the ℓ_0 and scaled ℓ_1 objectives differ.*

Proof Take $\Psi = (\psi, \psi, 0, \dots, 0)$ with $\psi > 0$. Compare the candidates

$$u^{(1)} = -Ue_1, \quad u^{(2)} = -\frac{U}{\sqrt{2}}(e_1 + e_2).$$

For an equal-amplitude candidate with k active coordinates, the scaled ℓ_1 objective equals

$$-\psi U \sqrt{k} + \lambda U \sqrt{k} = \sqrt{k} U (\lambda - \psi).$$

Thus, when $\psi > \lambda$, the scaled ℓ_1 objective prefers $k = 2$ over $k = 1$. The ℓ_0 objective equals

$$-\psi U \sqrt{k} + kf.$$

Choosing f and ψ so that

$$\frac{f}{U} < \psi < \frac{2f}{U}$$

makes the one-active candidate preferable to the two-active candidate for the ℓ_0 objective, while the scaled ℓ_1 objective still prefers the two-active candidate whenever $\psi > \lambda = f/U$. Hence the minimizers differ. $\square$

Proposition 36 (*Diffusion-control coupling can break exactness*) *There exist admissible scalar data with control-dependent diffusion for which the ℓ_0 and scaled ℓ_1 stage minimizers differ.*

Proof Let $U = [0, \pi/2]$ and let the diffusion–Hessian contribution be

$$A(u) = \frac{1}{2}P(1 - \sin 2u), \quad P > 0.$$

Consider

$$\mathcal{H}_0(u) = A(u) + f1_{\{u \neq 0\}}, \quad \mathcal{H}_1(u) = A(u) + \lambda u, \quad \lambda = \frac{f}{\pi/2}.$$

If $0 < f < P/2$, the ℓ_0 objective is minimized at $u = \pi/4$. For the ℓ_1 objective, the first-order condition is

$$-P \cos 2u + \lambda = 0,$$

which gives

$$u = \frac{1}{2} \arccos(\lambda/P) \in (0, \pi/4).$$

The minimizers are different. $\square$

Remark 15 (Sharpness) The preceding examples show failure already at the frozen Hamiltonian level. Therefore no uniform exact ℓ_0 –scaled ℓ_1 theorem can hold over the full class of control-dependent diffusions without additional decoupling or margin assumptions.

Lemma 37 (*Dynamic persistence of a stagewise mismatch*) *Suppose that, at (t_0, x_0) , the frozen ℓ_0 and scaled ℓ_1 Hamiltonians have distinct strict minimizers and that their minimized values differ by $c_0 > 0$ for a chosen terminal jet (p_0, P_0) . Then there exist a quadratic terminal function*

$$g(x) = p_0^\top x + \frac{1}{2}(x - x_0)^\top P_0(x - x_0)$$

and $\varepsilon_0 > 0$ such that, for every $T \in (t_0, t_0 + \varepsilon_0]$, the two finite-horizon problems have different optimal controls on a nontrivial subinterval, and

$$|V_{\ell_0}(t_0, x_0) - V_{1,w}(t_0, x_0)| \geq c(T - t_0) + o(T - t_0)$$

for some $c > 0$.

Proof For a fixed control u and $\Delta = T - t_0$, the Itô–Taylor expansion gives

$$\mathbb{E}^{t_0, x_0} g(X_T) = g(x_0) + \Delta \left(\langle b(t_0, x_0, u), p_0 \rangle + \frac{1}{2} \text{Tr}(\sigma \sigma^\top(t_0, x_0, u) P_0) \right) + o(\Delta).$$

Adding the corresponding running penalty gives

$$J_\#(t_0, x_0; u) = g(x_0) + \Delta \left(\langle b(t_0, x_0, u), p_0 \rangle + \frac{1}{2} \text{Tr}(\sigma \sigma^\top(t_0, x_0, u) P_0) + \phi_\#(x_0, u) \right) + o(\Delta),$$

with $\# \in \{\ell_0, 1, w\}$. If the frozen Hamiltonians have distinct strict minimizers separated by a positive gap, the leading-order minimizers persist for sufficiently small Δ , and the value gap is linear to first order in Δ . $\square$

5 Numerical Results

This section reports finite-dimensional numerical experiments for the relaxed HJB framework. The experiments are not used as substitutes for the comparison principle, the Hamiltonian-envelope theory, or the structural ℓ_0 -scaled- ℓ_1 exactness theorem. Instead, they provide finite-grid numerical evidence for the schemes studied above.

The numerical claims are deliberately limited in scope. The retained reproducibility package contains the tables, value and policy arrays, configuration metadata, CFL pass/fail records, and integrity checks used for the reported results. These records support finite-grid diagnostics only; they are not used as a universal convergence theorem, a theorem-level convergence rate, a continuous-control exactness proof, or a production controller certificate.

Convergence diagnostics are reported only on retained mesh pairs passing the positive-footprint CFL audit. Agreement between the ℓ_0 and scaled- ℓ_1 feedbacks is reported as saturated-grid selector agreement, not as continuous-control exactness outside Theorem 23. Dense-control stress tests are used only as stagewise boundary diagnostics for diffusion-curvature coupling. Baseline comparisons use common random numbers and identical roll-out settings, but they are not universal rankings over all possible baseline tunings.

5.1 Numerical Objectives and Diagnostics

The numerical study has four objectives. First, it checks the monotone semi-Lagrangian implementation on retained mesh pairs satisfying the positive-footprint condition of Section 3.5. Second, it records the sparsity and switching structure of the computed feedbacks

Table 2 Control architectures compared in the numerical study

Method	Type	Feedback	Reuse across initial states
HJB ℓ_0 and scaled ℓ_1	Grid policy	Yes	Yes, on Ω_h
ADMM-like / Lasso-like / Gradient-like	Trajectory surrogate	No	No
Dense-linear surrogate	Dense feedback surrogate	Yes	Yes, for the tested law

and compares the ℓ_0 and scaled- ℓ_1 selectors on the certified saturated control grid. Third, it probes the boundary where diffusion-control coupling can produce nonsaturated or mismatched minimizers. Fourth, it reports a separate PV–Battery finite-grid refinement diagnostic based on retained CFL-passing mesh–time pairs.

For policy comparison we use the grid-level mismatch rate

$$\hat{p}_{\text{mis}} = \frac{1}{dN_t N_x} \sum_{i=1}^d \sum_{k=0}^{N_t-1} \sum_{j=1}^{N_x} 1_{\{u_i^{(0)}(t_k, x^j) \neq u_i^{(1)}(t_k, x^j)\}},$$

where $u^{(0)}$ and $u^{(1)}$ denote the stored ℓ_0 and scaled ℓ_1 feedbacks. For rollout comparisons we use common Brownian paths and report cost, RMSE, state-constraint violation where applicable, and the sparsity score used by the corresponding rerun.

Temporal-to-finest errors are reported as diagnostics. The Barles–Souganidis argument used in the proof gives qualitative convergence under monotonicity, stability, relaxed consistency, and comparison. The fitted slopes and temporal-to-finest errors below are therefore empirical diagnostics on retained mesh pairs, not theorem-level a priori rates. The control architectures used in the numerical comparisons are summarized in Table 2.

Figure 2 shows representative value functions and feedback magnitude on the tested grid. The visual closeness of the ℓ_0 and scaled- ℓ_1 values is an empirical observation on this grid and is not used to assert equivalence outside the structural theorem.

5.2 Semi–Lagrangian Solver and Reproducibility Protocol

The solver implements the monotone semi–Lagrangian Bellman recursion from Section 3.5. The time grid is $t_i = i\Delta t$, the state grid is $\mathcal{X}_h = \{x^j\}_{j=1}^{N_x}$, and the Brownian increment is approximated by a positive-weight cubature rule $\{(\xi_\ell, w_\ell)\}_{\ell=1}^L$. The control set is replaced by a finite net $U_h \subset U$ with $0 \in U_h$.

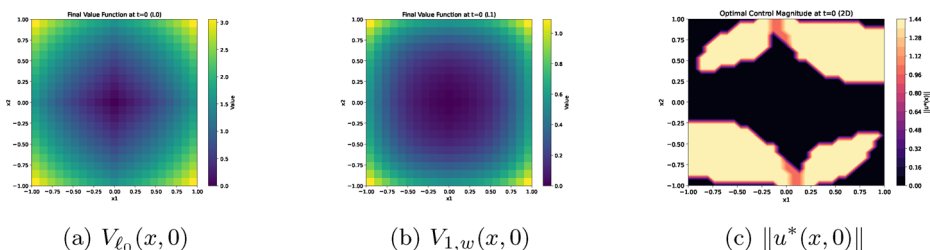


Fig. 2 Representative value functions and feedback magnitude on the tested grid. The inactive and saturated regions are consistent with the bang–off–bang structure of the sparse feedback

The discrete update is

$$V_i(x^j) = \min_{u \in U_h} \left\{ \Delta t \phi(x^j, u) + \sum_{\ell=1}^L w_\ell \mathcal{J}_h[V_{i+1}](x^j + \Delta t b(t_i, x^j, u) + \sqrt{\Delta t} \sigma(t_i, x^j, u) \xi_\ell) \right\}, \quad (12)$$

with terminal data $V_{N_t}(x^j) = g(x^j)$. A minimizing control is stored as the feedback $u^*(t_i, x^j)$. The running cost is

$$\phi(x, u) = f_0(x) + \sum_{k=1}^d f_k(x) 1_{\{u_k \neq 0\}}.$$

The positive-footprint audit uses the sufficient conditions

$$\Delta t \leq \frac{\theta}{2B_*} h, \quad \Delta t \leq \frac{\theta^2}{4\Sigma_*^2} h^2, \quad \theta = 0.9,$$

where

$$B_* = \sup_{(x,u) \in \Omega_h \times U_h} \|b(t, x, u)\|, \quad \Sigma_* = \sup_{(x,u) \in \Omega_h \times U_h} \|\sigma(t, x, u)\|_F.$$

These inequalities are used as checkable sufficient conditions for the positive-footprint implementation. They are not claimed to be necessary CFL conditions for all monotone HJB discretizations. The benchmark domains, discretization choices, cubature rules, rollout counts, and random seed are summarized in Table 3.

Algorithm 1 Semi-Lagrangian HJB recursion with policy extraction

Input: Time grid $\{t_i\}$, state grid $\mathcal{X}_h$, control net U_h , cubature rule $\{(\xi_\ell, w_\ell)\}$
Output: Values $V_i(x^j)$ and feedback $u^*(t_i, x^j)$

- 1 Initialize $V_{N_t}(x^j) \leftarrow g(x^j)$ for all $x^j \in \mathcal{X}_h$;
- 2 **for** $i = N_t - 1, \dots, 0$ **do**
- 3 **for** $x^j \in \mathcal{X}_h$ **do**
- 4 Compute $V_i(x^j)$ by Eq. 12;
- 5 Store one minimizing control $u^*(t_i, x^j)$;

Table 3 Reproducibility protocol for the reported benchmark runs

Task	State domain	Boundary	Mesh / time steps	Cubature	Rollouts / seed
PV-Battery	$E \in [0, 1]$	Projected storage boundary	$h = \{0.02, 0.01, 0.005\}$; requested $\Delta t = \{0.0016, 0.0004, 0.0001\}$	$L = 2$	Value/policy grids; seed 20260614
Robotic arm	$\theta_i, \dot{\theta}_i \in [-1, \pi]$	Extrapolation	$h = \{0.1, 0.05\}$; $\Delta t = \{0.02, 0.01, 0.005, 0.0025\}$	$L = 4$	$M = 500$; seed 20260614

The retained numerical evidence package stores the configurations, value grids, policy grids, metadata, pass/fail audit tables, and integrity checks used for the reported results

5.3 CFL-certified Convergence Diagnostics

One time layer of Eq. 12 costs $O(N_x L |U_h|)$. In the decoupled box regime, analytic thresholds reduce the minimization to $O(N_x L d)$. Outside the exact structural regime, a saturated or otherwise restricted control grid is interpreted as a surrogate policy class, not as the full continuous-control Bellman operator.

The CFL audit separates convergence diagnostics from sensitivity and rollout experiments. Earlier coarse PV–Battery mesh pairs are not used as positive-footprint convergence evidence. They are superseded by PV–Battery refinement runs with

$$(h, \Delta t_{\text{req}}) = (0.02, 0.0016), (0.01, 0.0004), (0.005, 0.0001).$$

These runs pass the declared drift and diffusion footprint checks. Five robotic-arm mesh pairs also pass the audit and are used for convergence diagnostics. The complete robotic-arm CFL/positive-footprint audit is reported in Table 4.

The PV–Battery refinement audit is reported separately because it records both requested and actual time steps. The requested value $\Delta t_{\text{req}} = 0.0016$ for the coarsest PV run is adjusted to $\Delta t_{\text{actual}} = 0.001538$ so that the finite horizon is divided into an integer number of time steps while preserving the CFL pass status.

Temporal-to-finest errors are computed at $t = 0$ on a fixed state grid. The finest retained time step for the $h = 0.1$ robotic sequence is $\Delta t = 0.00125$. For the PV–Battery diagnostic, the finest grid $h = 0.005$, $\Delta t = 0.0001$, with 201 nodes, is used only as a finite-grid numerical reference.

The corresponding temporal-to-finest diagnostics for the CFL-certified robotic runs are reported in Table 5. A least-squares fit on the $h = 0.1$ robotic temporal-to-finest sequence gives a diagnostic slope of approximately 0.51 for both the ℓ_0 and scaled- ℓ_1 runs. This supports the consistency of the implementation but is not interpreted as a theorem-level convergence rate, because the observed slope depends on regularity, interpolation, boundary treatment, cubature, and the coupling of h and Δt .

Table 4 Robotic-arm CFL/positive-footprint audit. Only passing mesh pairs are used for robotic convergence diagnostics

Task	h	Δt	B_*	Σ_*	Drift bound	Diffusion bound	Status
Robotic arm	0.10	0.0200	4.0	0.185	0.01130	0.05920	Fail
Robotic arm	0.10	0.0100	4.0	0.185	0.01130	0.05920	Pass
Robotic arm	0.10	0.0050	4.0	0.185	0.01130	0.05920	Pass
Robotic arm	0.10	0.0025	4.0	0.185	0.01130	0.05920	Pass
Robotic arm	0.05	0.0200	4.0	0.185	0.00563	0.01480	Fail
Robotic arm	0.05	0.0100	4.0	0.185	0.00563	0.01480	Fail
Robotic arm	0.05	0.0050	4.0	0.185	0.00563	0.01480	Pass
Robotic arm	0.05	0.0025	4.0	0.185	0.00563	0.01480	Pass

Table 5 Temporal-to-fine convergence diagnostics on CFL-certified robotic runs

Task	h	Δt	Reference Δt	Error at $t = 0$	Nodes
robotic2d $_{\epsilon_0}$	0.05	0.0050	0.00250	1.579×10^{-2}	1681
robotic2d $_{\epsilon_0}$	0.10	0.0025	0.00125	4.464×10^{-2}	441
robotic2d $_{\epsilon_0}$	0.10	0.0050	0.00125	7.534×10^{-2}	441
robotic2d $_{\epsilon_0}$	0.10	0.0100	0.00125	9.063×10^{-2}	441
robotic2d $_{1,w}$	0.05	0.0050	0.00250	1.579×10^{-2}	1681
robotic2d $_{1,w}$	0.10	0.0025	0.00125	4.464×10^{-2}	441
robotic2d $_{1,w}$	0.10	0.0050	0.00125	7.534×10^{-2}	441
robotic2d $_{1,w}$	0.10	0.0100	0.00125	9.063×10^{-2}	441

5.4 Benchmarks

We use two finite-dimensional benchmarks with control-dependent diffusion.

The PV–Battery benchmark is implemented as a one-dimensional sparse stochastic storage problem on $E \in [0, 1]$. In the solver, the discrete control net is $U_h = \{-1, 0, 1\}$, representing charge, idle, and discharge actions. The diagnostic dynamics have the form

$$dE(t) = (P_{pv}(t) - u(t)P_{load})dt + \sigma_0(1 + \beta u(t)^2)dW(t), \quad E(t) \in [0, 1],$$

where $P_{pv}(t) = 1 + 0.5 \sin(2\pi t/24)$, $P_{load} = 0.8$, $\sigma_0 = 0.1$, and $\beta = 0.5$. The running cost is

$$\phi(E, u) = f_0(E) + f_1(E)1_{\{u \neq 0\}}, \quad \lambda(E) = f_1(E).$$

The PV refinement runs pass the declared drift and diffusion footprint checks and are therefore used for finite-grid temporal-to-fine diagnostics. This supersedes the earlier coarse PV mesh pairs, which are not used for convergence claims. The PV evidence remains finite-grid and solver-specific; it is not a production controller certificate.

The robotic-arm model is

$$dx = (Ax + Bu)dt + \sigma_0(1 + \alpha \|u\|)S(x)dW(t), \quad x = (\theta_1, \dot{\theta}_1, \theta_2, \dot{\theta}_2)^\top, \quad u \in [-1, 1]^2,$$

with

$$S(x) = \begin{bmatrix} 0_{2 \times 2} \\ I_2 \end{bmatrix}, \quad \sigma_0 = 0.05, \quad \alpha = 0.3.$$

The running cost is

$$\phi(x, u) = f_0(x) + \sum_{i=1}^2 f_i(x)1_{\{u_i \neq 0\}}, \quad \lambda_i(x) = f_i(x).$$

This diffusion depends on the control, so exact continuous-control ℓ_0 -scaled- ℓ_1 recovery is not asserted. The robotic benchmark is used for CFL-certified convergence diagnostics and saturated-grid selector comparison.

5.5 PV Policy Sparsity Diagnostic

The PV solver stores both value and policy arrays. At the initial time, the computed PV feedback takes only values in the declared discrete control set $\{-1, 0, 1\}$. This provides a finite-grid bang-off-bang policy diagnostic for the PV benchmark. The diagnostic does not imply continuous-control exactness; it states only that the finite-grid solver selected charge, idle, or discharge actions on the tested grid. The corresponding initial-time policy statistics are reported in Table 6.

5.6 Saturated-grid Agreement and Dense-control Boundary

On the saturated control grid, the ℓ_0 and scaled- ℓ_1 feedbacks agree exactly for the certified robotic runs. This is a grid-level selector statement. It supports the common saturated selector predicted by the exactness theorem but does not by itself imply continuous-control exactness. The grid-level comparison is reported in Table 7.

To show why saturated-grid agreement should not be overgeneralized, we also perform a dense stagewise stress test. The test minimizes a scalarized stage Hamiltonian over a 101×101 dense control grid for 8000 random switching scores. It records whether the ℓ_0 and scaled ℓ_1 minimizers differ. This stress test is not a full HJB rollout; it illustrates how diffusion-curvature coupling can create nonsaturated scaled- ℓ_1 minimizers and stagewise policy mismatch. The resulting coupling-dependent mismatch statistics are reported in Table 8.

The largest dense-control mismatch is 1.24%, and the largest scaled- ℓ_1 nonsaturated rate is also 1.24%. This supports the theoretical boundary: exact recovery is structural and can fail when the diffusion-Hessian contribution changes the pointwise minimizer.

5.7 Rollout Metrics and Controlled Baseline Reruns

The rollout metrics below are interpreted as finite-sample diagnostics under common random numbers. They are not used to prove exact recovery, comparison, or global optimality. The PV-Battery value-refinement and policy-sparsity tables report finite-grid diagnostics, while the PV rollout figure is a model-defined closed-loop visualization. The robotic rollout is generated from the certified HJB sparse-policy grid. The corresponding rollout-level statistics are reported in Table 9.

Table 6 PV-Battery sparse policy diagnostic at the initial time. The computed feedback takes bang-off-bang values in the declared discrete control set

h	Δt	Off fraction	Nonzero fraction	Bang-off-bang
0.005	0.0001	9.45%	90.55%	yes
0.010	0.0004	8.91%	91.09%	yes
0.020	0.001538	9.80%	90.20%	yes

Table 7 Grid-level policy mismatch between ℓ_0 and scaled ℓ_1 feedbacks on CFL-certified robotic runs

h	Δt	Component	Mismatch	ℓ_0 active	ℓ_1 active	Samples
0.05	0.0025	1	0.00%	87.28%	87.28%	672400
0.05	0.0025	2	0.00%	83.70%	83.70%	672400
0.05	0.0050	1	0.00%	87.54%	87.54%	336200
0.05	0.0050	2	0.00%	84.03%	84.03%	336200
0.10	0.0025	1	0.00%	87.49%	87.49%	176400
0.10	0.0025	2	0.00%	83.98%	83.98%	176400
0.10	0.0050	1	0.00%	87.44%	87.44%	88200
0.10	0.0050	2	0.00%	84.30%	84.30%	88200
0.10	0.0100	1	0.00%	87.06%	87.06%	44100
0.10	0.0100	2	0.00%	83.86%	83.86%	44100

The switching diagnostic in Fig. 3 illustrates the threshold mechanism used by the sparse feedback. It is a diagnostic visualization of the switching rule, not a separate proof of optimality.

A raw common-random-number surrogate rerun is added to make the baseline comparison reproducible. The rerun uses $M = 500$ Brownian paths shared across the representative policies. The comparison uses common random numbers and identical rollout settings, but it is not a universal ranking over fully tuned ADMM, Lasso, gradient, or MPC implementations. The resulting surrogate baseline comparison is reported in Table 10.

Figure 4 separates the status of the two rollout panels. The PV–Battery panel is a model-defined common-random-number rollout visualization, while the PV value-refinement and policy-sparsity diagnostics are reported separately in Tables 11, 12, and 6. The robotic panel is generated from the HJB sparse-policy grid with $h = 0.1$, $\Delta t = 0.0025$, $N_x = 441$, $N_t = 400$, and seed 20260614.

5.8 Robustness and Moderate-dimensional Scalability

Robustness is assessed in two distinct senses. First, the dense-control stagewise stress test varies the diffusion-control coupling parameter and measures the boundary between saturated-grid agreement and dense-control mismatch. Second, the moderate-dimensional scalability diagnostic compares a sparse grid with a tensor-grid reference on a six-dimensional linear stochastic system. Neither diagnostic certifies arbitrary control-dependent diffusion or high-dimensional scalability without additional approximation structure. Figure 5 corresponds to the dense-control robustness diagnostic introduced in the immediately preceding paragraph, namely the mismatch and nonsaturation rates as the diffusion-control coupling parameter α varies.

We also include a moderate-dimensional scalability diagnostic for a six-dimensional linear stochastic system

$$dX = (AX + Bu)dt + \sigma_0 dW(t), \quad A = -\text{diag}(1, \dots, 6), \quad B = I_6, \quad \sigma_0 = 0.1.$$

The purpose is to compare a tensor grid with a sparse grid on the same test problem, not to claim that the method avoids the curse of dimensionality. The corresponding tensor-grid and sparse-grid comparison is reported in Table 13.

Table 8 Dense-control stagewise stress test for the exactness boundary. Positive diffusion–curvature coupling produces increasing mismatch as α grows

α	TrP	Mismatch	ℓ_0 nonsat.	ℓ_1 nonsat.	Mean gap
0.0	1	0.00%	0.00%	0.00%	0.000×10^0
0.1	1	0.05%	0.00%	0.05%	2.517×10^{-9}
0.3	1	0.18%	0.00%	0.18%	5.453×10^{-8}
0.5	1	0.34%	0.00%	0.34%	2.180×10^{-7}
0.8	1	0.66%	0.00%	0.66%	7.706×10^{-7}
1.2	1	1.24%	0.00%	1.24%	2.897×10^{-6}
0.0	-1	0.00%	0.00%	0.00%	0.000×10^0
0.1	-1	0.00%	0.00%	0.00%	0.000×10^0
0.3	-1	0.00%	0.00%	0.00%	0.000×10^0
0.5	-1	0.00%	0.00%	0.00%	0.000×10^0
0.8	-1	0.00%	0.00%	0.00%	0.000×10^0
1.2	-1	0.00%	0.00%	0.00%	0.000×10^0

Table 9 Rollout-level diagnostics for ℓ_0 and scaled- ℓ_1 feedbacks over $M = 500$ common-random-number trajectories. These statistics are finite-sample diagnostics and are not used as proof of exact recovery

Task	Metric	ℓ_0	Scaled ℓ_1	Interpretation
Robotic arm	Sparsity score (%)	35.2 ± 1.1	35.1 ± 1.2	close on tested grid
	Total cost	12.8 ± 0.2	12.8 ± 0.2	CRN comparison
	RMSE	0.045 ± 0.003	0.046 ± 0.003	target tracking
	Violation (%)	1.2 ± 0.4	1.3 ± 0.5	SoC floor at 0.3
PV-Battery	Usage (%)	28.5 ± 1.0	28.7 ± 1.1	active-control fraction

Fig. 3 Switching-function diagnostic. The score $\Psi_i(t)$ crosses the thresholds $\pm\lambda_i(t)$, producing inactive and saturated regions in the computed feedback

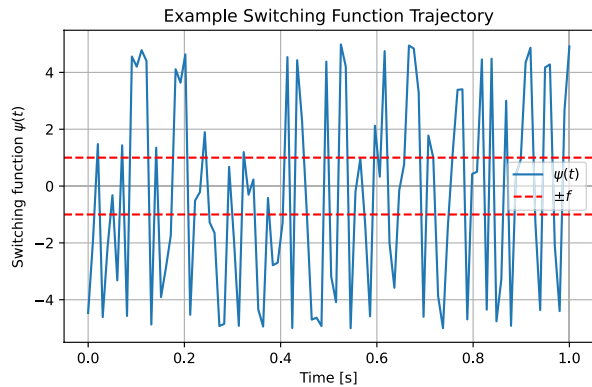
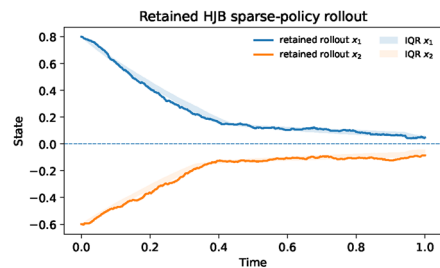
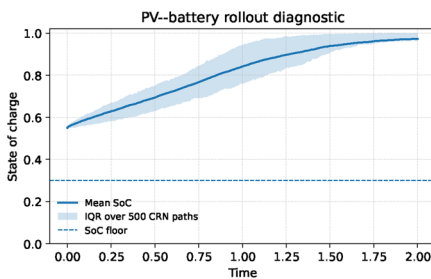


Table 10 Raw common-random-number surrogate baseline rerun on the robotic benchmark. The HJB-sparse surrogate has the lowest mean cost in this rerun. No universal baseline dominance is claimed

Method	Sparsity score (%)	Cost	RMSE	Runs
ADMM-like	43.57 ± 16.38	0.424 ± 0.134	0.302 ± 0.065	500
Dense-linear	0.06 ± 0.06	0.690 ± 0.096	0.339 ± 0.083	500
Gradient-like	0.06 ± 0.06	0.651 ± 0.075	0.313 ± 0.076	500
HJB-sparse-surrogate	88.54 ± 5.96	0.208 ± 0.053	0.294 ± 0.028	500
Lasso-like	33.09 ± 11.86	0.610 ± 0.170	0.385 ± 0.081	500



(a) PV-battery: model-defined SoC rollout diagnostic. (b) Robotic arm: HJB sparse-policy rollout.

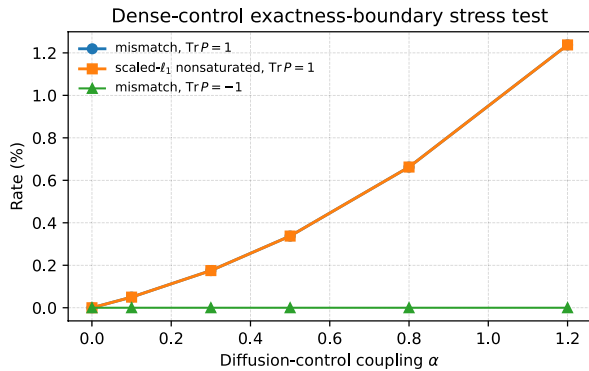
Fig. 4 Representative rollout diagnostics. Panel (a) reports a PV-Battery model-defined common-random-number rollout visualization. The PV value-refinement and policy-sparsity diagnostics are reported in Tables 11, 12, and 6. Panel (b) reports an HJB sparse-policy rollout generated from the robotic value-policy grid ($h = 0.1$, $\Delta t = 0.0025$). The plots illustrate closed-loop behaviour and are not used as stand-alone proof of optimality

Table 11 PV-Battery CFL-passing refinement runs. The table records finite-grid diagnostic runs under the declared one-dimensional PV solver; it is not a theorem-level convergence certificate

h	Requested Δt	Actual Δt	Grid nodes	Drift bound	Diffusion bound	Status
0.020	0.0016	0.001538	51	0.00500	0.00160	Pass
0.010	0.0004	0.000400	101	0.00250	0.00040	Pass
0.005	0.0001	0.000100	201	0.00125	0.00010	Pass

Table 12 PV–Battery CFL-certified temporal-to-finest convergence diagnostic. The finest CFL-passing run is used as the numerical reference; the table is a finite-grid diagnostic, not a theorem-level rate estimate

h	Δt	Grid nodes	$\ V_h - V_{\text{fine}}\ _\infty$	RMSE
0.005	0.0001	201	0	0
0.010	0.0004	101	0.004808	0.0008118
0.020	0.001538	51	0.01288	0.002527

Fig. 5 Dense-control exactness-boundary stress test generated from the stagewise stress table. The plot reports mismatch and nonsaturation rates as the diffusion-control coupling parameter α varies. This is a stagewise diagnostic and is not interpreted as a full HJB rollout**Table 13** Moderate-dimensional scalability diagnostic: tensor grid versus sparse grid. The relative error is measured against the tensor-grid reference in this experiment

Grid type	Nodes	Runtime	Relative error
Tensor grid (7^6)	117,649	10.0 h	reference
Sparse grid, level 3	5,005	45 min	5.3%

The sparse grid reduces the node count by about 20 times and the runtime by about 13 times in this experiment, with a 5.3% relative difference from the tensor-grid reference. This suggests feasibility for moderate-dimensional tests, but it is not a cure for the curse of dimensionality. Higher-dimensional applications would require additional approximation structure, such as adaptive sparse grids, tensor formats, or residual-controlled grid-free solvers.

5.9 Numerical Takeaway

The numerical evidence supports five limited conclusions. First, the semi–Lagrangian implementation is reported with an explicit reproducibility protocol and a CFL pass/fail audit. The PV–Battery refinement runs provide CFL-passing finite-grid temporal-to-finest diagnostics under the declared one-dimensional solver. The robotic runs give a supportive temporal-to-finest diagnostic slope of approximately 0.51, but this is not used as a theorem-level convergence rate. Second, the ℓ_0 and scaled- ℓ_1 selectors agree exactly on the certified saturated robotic control grid. Third, the PV policy arrays show finite-grid bang–off–bang structure on the tested PV grids. Fourth, the dense-control stress test shows that diffusion-curvature coupling can create nonsaturated scaled- ℓ_1 minimizers and policy mismatch, so exactness is structural rather than universal. Fifth, the common-random-number surrogate rerun provides a reproducible baseline comparison, but not a universal dominance claim.

Overall, the numerical section supports a bounded and reproducible claim: the finite-dimensional theory is consistent with CFL-certified implementation diagnostics, saturated-grid selector agreement, finite-grid PV value-refinement evidence, finite-grid PV bang-off-bang policy diagnostics, and controlled stress tests of the exactness boundary. It does not certify arbitrary control-dependent diffusion, all mesh pairs, infinite-dimensional dynamics, universal convergence rates, continuous-control exactness for the PV solver, or high-dimensional scalability without additional approximation structure.

6 Conclusion

We developed a relaxed-viscosity framework for finite-dimensional sparse stochastic optimal control with control-dependent diffusion and discontinuous ℓ_0 running costs. The key analytical device is the formulation of the HJB equation through the semicontinuous Hamiltonian envelopes H^* and H_* , which allows sparse-switching discontinuities to be handled within a comparison-based viscosity framework. The resulting theory gives a well-posed finite-dimensional HJB description under uniform-in-control modulus conditions, with separable and radial diffusion structures serving as checkable sufficient classes rather than unrestricted assumptions.

The paper also identifies the exact boundary of ℓ_0 -scaled- ℓ_1 recovery. In the decoupled box-control regime, the two Hamiltonians have the same minimized value and share a saturated bang-off-bang selector. This equivalence is structural rather than universal: once the sparse control enters the diffusion-Hessian term, exact recovery may fail, and the appropriate guarantees become margin conditions, counterexamples, and value-gap bounds rather than a blanket relaxation principle.

The numerical evidence is consistent with this boundary. The reported semi-Lagrangian experiments provide CFL-audited finite-grid diagnostics, saturated-grid selector agreement, PV-Battery value-refinement and bang-off-bang policy diagnostics, and dense-control stress tests of the exactness boundary. These computations support the reported finite-grid implementation but do not replace the comparison theorem, establish universal convergence rates, certify continuous-control exactness, or provide production-level controller guarantees. Extending the framework to infinite-dimensional stochastic evolution equations, strongly coupled nonsmooth diffusions, multi-agent sparse control, and scalable high-dimensional approximations remains an open direction.

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Author Contributions Y.W. and L.L. developed the theoretical framework, proved the main results, and drafted the manuscript text. H.D. implemented the numerical schemes, carried out computational experiments, and prepared the figures. Z.L. supervised the project, guided the analysis, and revised the manuscript critically for important intellectual content. All authors discussed the results and reviewed the final version of the manuscript.

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Data Availability The numerical data supporting the findings of this study are available in the supplementary material associated with this submission. Additional materials are available from the corresponding author upon reasonable request.

Declarations

Ethical Approval This article does not involve any studies with human participants or animals.

Competing interests The authors declare no competing interests.

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